HPSC FSO Master Handout: Modern Food Packaging, Migration Regulations & Instrumental Food Analysis

0. Orientation & Exam Strategy

HPSC Food Safety Officer (Group-B) Master Study Handout

Module: Modern Food Packaging Technology, Migration Regulations & Instrumental Food Analysis

Prepared in the voice of a Senior Professor of Food Packaging Technology and Instrumental Analytical Chemistry, acting as Chief Paper Setter. Every section is written from first principles (polymer physics, material chemistry, spectroscopy, chromatography), then anchored to the statutory number (limit, schedule, prohibition), and finally converted into the examination deliverable (trap box, count-based pair, descriptive skeleton).

How the HPSC pattern rewards you

Stage	Marks	Format	Share of final merit	What is actually tested
Stage 1: Objective Screening Test	100	MCQs, 0.25 negative marking per wrong answer	12.5 percent	Precision recall of numbers, names of schedules, polymer properties, detector selectivity; heavy use of "how many of the following statements are correct" (count-based pairs) and absolute qualifiers (always, never, only, all, none)
Stage 2: Subject Knowledge Test	150	Conventional descriptive answers, 3 hours	87.5 percent	Structured answer writing: definition, mechanism, labelled schematic, comparative table, statutory citation, conclusion

Rules of engagement for Stage 1

1. Never answer a count-based pair by instinct. Write c1, c2, c3 in the margin for each statement (1 if correct, 0 if wrong), then compute N = c1 + c2 + c3 and match it to the option. This single discipline removes the majority of negative-marking losses.
2. Treat every absolute qualifier as a red flag. "EVOH is always the best barrier" is wrong because EVOH fails at high humidity. "Recycled plastics are completely banned" is wrong after the FSSAI certified food-grade recycled PET exception.
3. Distinguish look-alike pairs on sight: PVC versus PVDC, LDPE versus LLDPE, FAAS versus GFAAS, ECD versus FID, C-enamel versus R-enamel, overall migration versus specific migration, active versus intelligent packaging, time-temperature indicator versus freshness indicator.

Rules of engagement for Stage 2

1. Open with a one-line definition or statutory citation (regulation name, year, schedule).
2. Give a mechanism (what happens at the molecular or instrumental level) before giving the application.
3. Add one labelled schematic (described in words if you cannot draw) and one comparative table; examiners award marks for structure.
4. Close with a regulatory or public-health linkage, since the post is Food Safety Officer, not Food Technologist.

The thirty numbers you must never forget

Number	Fact
60 mg/kg or 10 mg/dm^2	Overall migration limit for plastics (FSS Packaging Regulations, 2018, Schedule IV; same as EU Regulation 10/2011)
6 dm^2 per kg	EU convention: 1 kg of food is assumed to contact 6 dm^2 of packaging, so 10 mg/dm^2 × 6 = 60 mg/kg
1.5 mg/kg	Specific migration limit for DEHP (bis(2-ethylhexyl) phthalate)
0.04 mg/kg	Specific migration limit for antimony (PET polymerisation catalyst residue)
24 December 2018; 1 July 2019	FSS (Packaging) Regulations notified; enforced
15 µm	Aluminium foil below this thickness is prone to pinholes and must be laminated
9 to 12 µm	Foil gauge used inside retort pouches
121 °C, 20 to 40 min	Typical retort pouch sterilisation
0.910 to 0.925; 0.941 to 0.965 g/cm^3	LDPE density; HDPE density
1.33 to 1.40 g/cm^3	PET density (sinks in water, basis of float-sink separation from polyolefins)
1.6 to 1.7 g/cm^3	PVDC, the densest common packaging polymer
27 to 48 mol percent	Ethylene content range of commercial EVOH
2.8 to 11.2 g/m^2	Electrolytic tin coating mass on tinplate
250 mg/kg	FSSAI limit for tin in canned foods
550 °C	Muffle furnace temperature for total ash
6.25	General nitrogen-to-protein factor (assumes 16 percent nitrogen in protein)
6.38 and 5.70	Factors for milk and for wheat

Number	Fact
1.25 percent H2SO4 then 1.25 percent NaOH, 30 min each	Crude fibre digestion
40 to 60 °C	Boiling range of petroleum ether used in Soxhlet
253.7 nm	Mercury resonance line used in cold vapour AAS
6000 to 10000 K	Argon plasma temperature in ICP-MS
27.12 MHz	Usual RF frequency of the ICP load coil
15.76 eV	First ionisation energy of argon; elements below this ionise efficiently in the plasma
70 eV	Standard electron ionisation energy in GC-MS
63Ni	Beta source of the electron capture detector
365 nm excitation, about 435 to 450 nm emission	Fluorescence detection of aflatoxins
0.5 µg/kg	FSSAI limit for aflatoxin M1 in milk
0.01 mg/kg	FSSAI default maximum residue limit for pesticides not specifically listed
2 percent	FSSAI ceiling on trans fatty acids in fats and oils (from 1 January 2022)
0.25	Negative marking per wrong answer in HPSC Stage 1

...is quantified by HPLC after pre-column derivatisation, because histamine itself has no useful chromophore: dansyl chloride gives a UV-absorbing (254 nm) derivative, while o-phthalaldehyde (OPA) with a thiol gives a fluorescent isoindole derivative (excitation 340 nm, emission 450 nm) read on a fluorescence detector at microgram-per-kilogram sensitivity. The FSSAI limit for histamine in fish and fishery products is 200 mg/kg (the Codex guideline is 100 mg/kg as the average of nine sample units, with 200 mg/kg as the hygiene limit for any single unit). The hazard is scombroid poisoning: temperature-abused tuna, mackerel, sardine and mahi-mahi allow bacterial histidine decarboxylase (Morganella morganii, Klebsiella, Hafnia) to convert free histidine to histamine, which is heat stable and survives canning and cooking, so the Food Safety Officer must enforce time-temperature control at landing rather than rely on processing.

4.1.6 Remaining HPLC notes: separation science you must be able to write

- Reversed-phase versus normal-phase: in reversed-phase HPLC the stationary phase is non-polar (octadecylsilane, C18, bonded to 3 to 5 µm silica) and the mobile phase is polar (water with methanol or acetonitrile, often buffered); polar analytes elute first, and increasing the organic fraction shortens retention. Normal-phase HPLC uses bare silica or amino-bonded silica with a non-polar mobile phase (hexane with isopropanol); non-polar analytes elute first. About 80 percent of food methods are reversed-phase.
- Isocratic versus gradient elution: isocratic keeps mobile-phase composition constant (simple, compatible with the refractive index detector); gradient elution raises the organic fraction during the run to elute strongly retained compounds with sharp peaks (required for mixed vitamin or multi-colour screens, incompatible with the refractive index detector because the baseline drifts with composition).
- Column efficiency: the van Deemter equation $H = A + B/u + C \cdot u$ relates plate height H to linear velocity u; A is eddy diffusion (particle size and packing), B is longitudinal diffusion, C is resistance to mass transfer. Smaller particles reduce A and C, which is why ultra-high-performance liquid chromatography (UHPLC) uses sub-2 µm particles at pressures up to 1000 to 1500 bar to give ten times faster runs with higher resolution.

- Guard columns protect the analytical column from matrix; in-line degassing prevents pump cavitation; sample clean-up uses solid-phase extraction (C18 or immunoaffinity cartridges for aflatoxins) and syringe filtration at 0.45 µm.

Detector	Principle	Selectivity	Typical sensitivity	Gradient compatible	Food applications
UV-Vis (variable wavelength)	Beer-Lambert absorbance at a set wavelength	Compounds with chromophores	ng range	Yes	Preservatives (benzoate, sorbate at 230 to 254 nm), caffeine (272 nm), sweeteners, colours (visible region)
Photodiode array (PDA, DAD)	Full spectrum 190 to 800 nm recorded continuously	As UV-Vis, plus spectral confirmation and peak-purity check	ng range	Yes	Synthetic colours (each dye identified by its absorption maximum: tartrazine 427 nm, sunset yellow 482 nm, carmoisine 516 nm, brilliant blue 630 nm), vitamins, flavonoids
Fluorescence (FLD)	Excitation at one wavelength, emission at a longer wavelength	Only fluorescent molecules or fluorescent derivatives	pg range, 10 to 1000 times more sensitive than UV	Yes	Aflatoxins (365 nm excitation, 435 to 450 nm emission), ochratoxin A, histamine after OPA, vitamin B1 after thiochrome oxidation, vitamin B2, polycyclic aromatic hydrocarbons
Refractive index (RID)	Difference in refractive index between eluent and reference cell	Universal for non-UV compounds	µg range (low)	No (isocratic only); temperature sensitive	Sugar profiling (glucose, fructose, sucrose, lactose, maltose), sugar alcohols
Evaporative light scattering (ELSD)	Eluent nebulised, solvent evaporated, particles scatter light	Universal for non-volatile analytes	ng to µg	Yes	Sugars, lipids, triglycerides, saponins
Electrochemical (amperometric)	Oxidation or reduction at a working electrode	Electroactive compounds	pg	Limited	Vitamin C, catecholamines, pulsed amperometric detection of sugars on anion-exchange columns

Detector	Principle	Selectivity	Typical sensitivity	Gradient compatible	Food applications
Mass spectrometer	Mass-to-charge ratio of ions	Universal and structurally specific	pg to fg	Yes	Multi-residue pesticides, veterinary drugs, mycotoxins, melamine, acrylamide (Section 4.3)

4.1.7 Core HPLC applications in FSSAI enforcement

1. Vitamins: fat-soluble vitamins A (retinol, 325 nm), D (265 nm), E (tocopherols, 292 nm UV or fluorescence) and K after saponification and hexane extraction, reversed-phase C18 with methanol; water-soluble vitamins (B1, B2, B3, B6, B9, C) by ion-pair reversed-phase with UV or fluorescence detection; vitamin C by reversed-phase on a C18 column at 245 nm with metaphosphoric acid stabilisation. Needed to verify fortification claims (FSS (Fortification of Foods) Regulations, 2018, for example vitamin A and D in edible oil and milk, iron, folic acid and B12 in wheat flour and rice).
2. Synthetic food colours: extraction of the dye on wool or on a polyamide or C18 solid-phase cartridge, reversed-phase gradient with ammonium acetate buffer and methanol, PDA detection. India permits only eight synthetic colours (tartrazine, sunset yellow FCF, carmoisine, ponceau 4R, erythrosine, indigo carmine, brilliant blue FCF, fast green FCF) at a maximum of 100 mg/kg singly or in combination (200 mg/kg in a few categories), so the method must separate and identify non-permitted dyes such as metanil yellow, rhodamine B, orange II, auramine and Sudan dyes, which are a prosecutable offence.
3. Aflatoxins B1, B2, G1 and G2: extraction with methanol-water, immunoaffinity column clean-up, reversed-phase C18 with water-methanol-acetonitrile, fluorescence detection. Aflatoxins B1 and G1 fluoresce weakly in aqueous mobile phases, so post-column derivatisation is applied: electrochemically generated bromine (Kobra cell), iodine solution added by a second pump, trifluoroacetic acid pre-column, or a photochemical reactor (UV lamp) that hydrates the terminal furan double bond into the strongly fluorescent hemiacetal. FSSAI limits (Contaminants, Toxins and Residues Regulations, 2011, as amended): total aflatoxins 15 µg/kg and aflatoxin B1 10 µg/kg in cereals, nuts and oilseeds, 30 µg/kg total in certain spices such as chillies and nutmeg, and aflatoxin M1 0.5 µg/kg in milk.
4. Sugar profiling: aminopropyl (NH2) column, acetonitrile-water 75:25 isocratic, refractive index detection; separation order fructose, glucose, sucrose, maltose, lactose; used to detect added sucrose in honey and fruit juices and to verify sugar-free claims. High-performance anion-exchange chromatography with pulsed amperometric detection gives higher sensitivity.
5. Other routine targets: preservatives (benzoate, sorbate, propionate), artificial sweeteners (saccharin, aspartame, acesulfame-K, sucralose by ELSD), caffeine in energy drinks (FSSAI ceiling 300 mg/L), melamine screening, polyphenols and curcumin content in turmeric, capsaicin in chilli, patulin in apple juice (50 µg/kg limit).

HPSC EXAM TRAP: detector matching
Refractive index detection is universal but insensitive and cannot be used with gradient elution; fluorescence is the most sensitive but only for fluorescent molecules or their derivatives; the photodiode array is the detector that confirms identity by spectrum. Aflatoxins are measured by fluorescence after post-column derivatisation, not by refractive index; sugars are measured by refractive index, not by UV (they have no chromophore).

Part 1. Packaging Materials & Barrier Properties

PART 1: FOOD PACKAGING MATERIALS AND BARRIER PROPERTIES

1.0 Why packaging exists: the four functions

1. Containment: holding the product in a defined unit.
2. Protection: against physical (shock, compression), chemical (oxygen, moisture, light, odour) and biological (microbes, insects, rodents) hazards; this is where barrier science lives.
3. Convenience: portioning, opening, reclosing, microwaveability, retortability.
4. Communication: labelling, traceability, tamper evidence; this is where the Food Safety Officer meets the Labelling and Display Regulations, 2020.

1.1 Barrier science from first principles

A polymer film is not a wall; it is a solvent. Gas or vapour transport follows the solution-diffusion model:

1. The permeant dissolves in the polymer surface (governed by the solubility coefficient S).
2. It diffuses down a concentration gradient (governed by the diffusion coefficient D).
3. It desorbs on the other side.

Permeability coefficient P = D × S, and the steady-state transmission rate is proportional to P × area × partial-pressure difference divided by thickness. Everything in packaging barrier design manipulates D or S.

Factor	Effect on barrier	Reason
Crystallinity	Higher crystallinity gives lower permeability	Crystallites are impermeable; permeant must take a tortuous path through amorphous regions (HDPE beats LDPE)
Polarity match	Polar polymers (EVOH, nylon) resist non-polar oxygen but absorb water; non-polar polyolefins resist water but pass oxygen	"Like dissolves like": solubility S is high when permeant and polymer have similar polarity
Orientation	Biaxial orientation (BOPP, BOPET, OPA) improves barrier, stiffness and clarity	Strain-induced crystallisation and chain alignment
Temperature	Permeability rises with temperature	Arrhenius relationship; retort conditions temporarily open the structure
Relative humidity	Hydrophilic polymers (EVOH, nylon, cellophane) lose barrier at high RH	Water plasticises the hydrogen-bonded network, increasing free volume and D
Thickness	Transmission rate inversely proportional to thickness	Longer diffusion path
Metallisation or foil	Approaches zero transmission	Inorganic layers have essentially no free volume; limited only by pinholes and cracks

Standard units and test methods:

- WVTR (water vapour transmission rate): g/m²·day at 38 °C and 90 percent RH (ASTM E96 gravimetric cup; ASTM F1249 infrared sensor).
- OTR (oxygen transmission rate): cm³/m²·day·atm at 23 °C, usually 0 or 50 percent RH (ASTM D3985 coulometric sensor). EVOH must always be quoted with its RH because its OTR can rise ten to a hundred fold at 90 percent RH.

HPSC EXAM TRAP: the barrier paradox
No single homopolymer is excellent against both oxygen and water. Polyolefins (PE, PP) are good moisture barriers and poor oxygen barriers; EVOH and nylon are the reverse. PVDC is the only common polymer that is good against both, and aluminium foil is the only material that is essentially total against both plus light. Any statement that EVOH or nylon is a "good moisture barrier" is wrong.

1.2 Glass

Composition (soda-lime-silica container glass)

Constituent	Raw material	Approximate percent	Role
Silica, SiO2	Sand	70 to 74	Network former; gives rigidity and chemical inertness
Soda, Na2O	Soda ash, Na2CO3	12 to 16	Flux; lowers melting temperature from about 1700 °C to about 1500 °C
Lime, CaO	Limestone, CaCO3	10 to 13	Stabiliser; without it soda glass is water-soluble (water glass)
Alumina, Al2O3	Feldspar	1 to 3	Chemical durability, reduces devitrification
Cullet (recycled glass)	Crushed glass	30 to 60 of batch	Lowers melting energy; infinitely recyclable
Colourants	Iron oxides and chromium (green); carbon and sulfur (amber)	trace	Amber absorbs ultraviolet below about 450 nm, protecting beer, edible oil and milk from light-induced oxidation

Properties

- Chemical inertness: attacked only by hydrofluoric acid and hot concentrated alkali; no migration of organic substances, which is why glass is the reference inert material. Minor alkali (sodium) leaching into acidic or pharmaceutical products is controlled by hydrolytic resistance classes.
- Absolute barrier to gases, vapours, micro-organisms and odours; transparent (consumer appeal but light sensitivity); rigid, stackable, reclosable, microwave compatible, fully recyclable.
- Drawbacks: heavy (density about 2.5 g/cm³), brittle, fracture hazard (glass fragments are a critical physical hazard in HACCP plans), poor thermal shock resistance.

Thermal shock resistance

Glass fails under sudden temperature change because the outer surface contracts or expands before the interior, generating tensile stress. Soda-lime container glass survives a sudden differential of only about 40 to 50 °C (hot filling into a cold jar, or cold water on a hot jar, is a classic breakage cause), which is why hot-fill lines pre-warm containers and retort processes raise and lower temperature gradually. Borosilicate

glass (about 12 percent B2O3; laboratory and ovenware) has a coefficient of thermal expansion of about 3.3 × 10^-6 per kelvin versus about 9 × 10^-6 for soda-lime glass, so it tolerates differentials well above 100 °C.

Annealing

After forming (blow-and-blow for narrow-neck bottles, press-and-blow for jars on the IS machine) the container cools unevenly and locks in internal stresses. It passes through an annealing lehr held at about 550 to 600 °C (near the annealing point, above the strain point) and is then cooled slowly under control. Un-annealed glass can shatter spontaneously or under minimal impact. Annealing quality is checked with a polariscope (strain viewer).

Surface coatings and treatments

- Hot-end coating: tin tetrachloride or titanium tetrachloride vapour applied at about 500 °C, forming a thin SnO2 or TiO2 film that heals surface micro-cracks and anchors the cold-end coating.
- Cold-end coating: polyethylene wax or stearate emulsion applied at about 100 °C for lubricity, so bottles do not scratch each other on filling lines (surface scratches cut glass strength by up to 60 percent).
- Interior surface treatment (dealkalisation): sulfur dioxide or ammonium sulfate introduced at the hot end converts surface sodium to sodium sulfate, which is washed away, leaving a silica-rich inner skin with improved hydrolytic resistance. This is the "interior coating" concept in the syllabus; it reduces alkali extraction into acidic beverages and pharmaceuticals.
- Closures: crown cork (beer, carbonated drinks), lug cap with plastisol liner (vacuum-sealed jars; the safety button indicates loss of vacuum), roll-on pilfer-proof (ROPP) aluminium caps.

Statutory note: FSS (Packaging) Regulations, 2018, Schedule II covers glass containers, requiring conformity to the listed Indian Standards and freedom from defects that could lead to breakage or fragments.

1.3 Metal packaging

1.3.1 Tinplate

Tinplate is low-carbon mild steel (about 0.15 to 0.5 mm) electrolytically coated with tin. From the steel outwards the structure is:

1. Steel base: strength and formability.
2. Iron-tin alloy layer (FeSn2): formed during reflow melting of the electrodeposited tin; improves corrosion resistance.
3. Free tin layer: sacrificial and corrosion resistant.
4. Passivation film: chromium oxide produced by cathodic treatment in sodium dichromate solution (the common "311" passivation); prevents tin oxide growth and sulfide staining, and improves lacquer adhesion.
5. Oil film: dioctyl sebacate (DOS) or similar, a few mg/m², for lubrication during handling and forming.

Electrolytic tinplate (ETP) replaced hot-dipped tinplate from the 1930s and allows differential coating (different tin mass on inside and outside). Coating mass ranges from 2.8 to 11.2 g/m² per side (grades such as E2.8/2.8, E5.6/2.8, E11.2/5.6). Heavier tin is used for unlacquered (plain) cans of acidic fruit products.

Why plain tin cans work for some fruits: in de-aerated acidic fruit packs tin is anodic to steel, so tin dissolves slowly and sacrificially, protecting the steel from pitting; the slight tin pickup is harmless below the FSSAI limit of 250 mg/kg and even preserves colour by reducing anthocyanin oxidation. In the presence of residual oxygen, nitrates or certain pigments the polarity can reverse and rapid detinning or hydrogen swell follows.

Can corrosion and spoilage vocabulary: hydrogen swell (internal corrosion liberates hydrogen; "hydrogen springer"), flipper, springer, soft swell, hard swell (the last three progressively indicate gas from microbial spoilage), sulfide black (iron sulfide staining), and pinholing.

Lacquers and enamels (internal coatings)

Enamel	Chemistry	Used for	Not used for	Reason
R-enamel (regular, oleoresinous)	Natural resins and drying oils, baked	Acid fruits, fruit juices, tomato products, pickles	Sulfur-bearing foods	Provides acid resistance but allows sulfide staining
C-enamel (corn enamel)	Oleoresinous resin pigmented with zinc oxide	Sweet corn, peas, beans, meat, fish, seafood (protein foods that release hydrogen sulfide on heating)	Acid fruits	Zinc oxide reacts with H2S to form white zinc sulfide instead of black iron or tin sulfide; in acid packs zinc would be extracted
Epoxy-phenolic	Bisphenol A epoxy cross-linked with phenolic resin	General purpose: vegetables, meat, beer, beverages	Where BPA-free is demanded	Excellent adhesion and chemical resistance; BPA migration concern (see Part 3)
Vinyl (organosol)	PVC dispersion, often over epoxy base coat	Beer, soft drinks, flavour-sensitive products	High-temperature retorting	Flavour inert but limited heat resistance
Phenolic	Phenol-formaldehyde	Meat, fish	Flexible deep-draw applications	High chemical resistance, brittle
Acrylic	Acrylic resins, often white pigmented	Vegetables, white interior cans	Strongly aggressive products	Good appearance, moderate resistance
Polyester, polyolefin, BPA-NI coatings	Non-epoxy chemistries	Replacement for epoxy where BPA restrictions apply		Driven by EU restriction of BPA in food contact materials

HPSC EXAM TRAP: C-enamel
C-enamel is chosen to prevent black sulfide discolouration in sulfur-rich protein foods (corn, peas, meat, fish). It is not chosen for "better corrosion resistance in acid fruits"; that is R-enamel or heavy-tin plain cans. If a question pairs "C-enamel: acid fruits", mark it wrong.

Can construction: three-piece cans (body with welded side seam plus two ends; lead solder is banned) and two-piece cans (drawn and wall-ironed, DWI, for beverages; drawn and redrawn, DRD, for food). The double seam (body hook, cover hook, overlap, sealing compound) is the hermetic closure and is inspected by seam tear-down (overlap, tightness rating). Tin-free steel (electrolytically chromium-coated steel, ECCS) carries chromium and chromium oxide instead of tin; it must be lacquered on both sides, cannot be soldered, and is used for ends and for welded bodies.

1.3.2 Aluminium

- Density 2.7 g/cm³ versus 7.8 for steel: lightweight, lower transport energy; alloy 3004 for can bodies, 5182 for easy-open ends.
- Spontaneous passive Al2O3 film gives corrosion resistance in neutral media and immunity to sulfide staining; however aluminium is attacked by acids and alkalis and by chloride (pitting), so food cans are always lacquered (epoxy or polyester).

- Two-piece DWI beverage cans with stay-on-tab easy-open ends; non-magnetic, so separation at recycling uses eddy-current separators.
- Aluminium foil: rolled to 6 to 200 µm; food foil is annealed (soft temper); it has dead-fold (keeps a crease), is opaque (total light barrier), heat-stable, and when pinhole-free offers essentially zero OTR and WVTR.

The foil thickness to pinhole relationship: pinholes arise from rolling inclusions and double rolling of thin gauges. Below about 15 µm (some references say 17 µm) the pinhole count rises steeply; a 9 µm foil may carry several hundred pinholes per square metre, a 7 µm foil even more. Hence:

1. Foil above 15 µm (for example 20 µm semi-rigid trays, household foil) can be used alone as a near-absolute barrier.
2. Foil of 6 to 12 µm is always laminated to plastic films (PET on the outside, polyolefin on the inside); the polymer layers bridge pinholes so the laminate behaves as a near-total barrier even though the foil itself is not.
3. The retort pouch uses 9 to 12 µm foil for exactly this reason.

HPSC EXAM TRAP: foil and "100 percent barrier"
A thin foil is a 100 percent barrier only in the laminate, not on its own. The correct statement is: "aluminium foil below 15 µm is prone to pinholes and is therefore laminated".

1.4 Food polymers and plastics

Master comparative table

Polymer (recycling code)	Chemistry	Density, g/cm³	Oxygen barrier	Water vapour barrier	Melting or softening	Typical food uses
LDPE (4)	High-pressure free-radical polyethylene, long-chain branched, 40 to 50 percent crystalline	0.910 to 0.925	Poor	Good	Tm 105 to 115 °C	Bread bags, frozen food bags, heat-seal layer, squeeze bottles, milk pouches (blend)
LLDPE (4)	Ziegler-Natta or metallocene copolymer with butene, hexene or octene; short-chain branches	0.915 to 0.925	Poor	Good	Tm 120 to 125 °C	Stretch wrap, heavy-duty sacks, tougher seal layers, Indian milk pouches (LLDPE blends)
HDPE (2)	Low-pressure Ziegler-Natta, linear, 70 to 90 percent crystalline	0.941 to 0.965	Poor to moderate	Very good (about four times better than LDPE)	Tm 130 to 135 °C	Milk jugs, edible oil cans, crates, caps, boil-in-bag
PP (5)	Isotactic polypropylene; BOPP is biaxially oriented; CPP is cast	0.90 to 0.91 (lightest)	Poor	Very good	Tm 160 to 170 °C (retortable, hot-fillable)	Yoghurt cups, retort trays, caps, BOPP snack films, CPP retort seal layer

Polymer (recycling code)	Chemistry	Density, g/cm³	Oxygen barrier	Water vapour barrier	Melting or softening	Typical food uses
PET (1)	Polyester of terephthalic acid and ethylene glycol; stretch-blow moulded	1.33 to 1.40	Good to moderate	Moderate	Tg about 75 °C, Tm 250 to 260 °C	Carbonated beverage and water bottles, edible oil, ovenable CPET trays, BOPET outer layer of laminates
PVC (3)	Polyvinyl chloride, amorphous, 57 percent chlorine; plasticised or unplasticised	1.35 to 1.45	Moderate	Moderate (poor when plasticised)	Tg about 80 °C	uPVC mineral water and oil bottles, cling film (plasticised), blister packs, lid gaskets
PVDC (7 or 3)	Polyvinylidene chloride copolymer (Saran), highly crystalline, high chlorine	1.6 to 1.7 (densest)	Excellent	Excellent	Softens about 150 °C, degrades easily	Barrier coating on cellophane, BOPP and PET; cheese and meat shrink films; chub packs
EVOH (7)	Hydrolysed ethylene-vinyl acetate copolymer, 27 to 48 mol percent ethylene	1.12 to 1.20	Exceptional when dry	Poor (hygroscopic)	Tm 160 to 190 °C	Core layer in co-extruded ketchup bottles, retort trays, cheese and meat films
Nylon 6, OPA (7)	Polyamide from caprolactam; OPA biaxially oriented	1.13 to 1.14	Good when dry	Poor	Tm 215 to 220 °C	Puncture-resistant layer in retort pouches, vacuum bags, thermoformed webs
PS, EPS (6)	Polystyrene; expanded with pentane	1.04 to 1.06	Very poor	Poor	Tg about 100 °C	Cups, trays, foam insulation; styrene migration concern
PC (7)	Polycarbonate from bisphenol A	1.20	Poor	Poor	Tg about 150 °C	Water dispensers, formerly baby bottles; BPA migration concern

1.4.1 Polyethylene family: LDPE, LLDPE, HDPE

All three are chains of CH2 units; the difference is branching, which controls crystallinity, which controls density, stiffness, melting point and barrier.

- LDPE is made at 1000 to 3000 bar by free-radical polymerisation; backbiting creates long branches that prevent close packing. Result: low density, low crystallinity, excellent clarity, flexibility and the best

heat-sealability (wide sealing window, seals through light contamination), but poor oxygen, aroma and grease barrier.

- LLDPE is made at low pressure with Ziegler-Natta or metallocene catalysts by copolymerising ethylene with an alpha-olefin, so branches are short and uniform. It has similar density to LDPE but higher tensile, puncture and tear strength and a higher melting point, giving down-gauging (thinner films with the same strength) and hot-tack strength for form-fill-seal machines. It is slightly hazier and harder to process than LDPE, so LDPE and LLDPE are commonly blended.
- HDPE is linear with almost no branching; chains pack tightly, giving the highest density, stiffness, melting point and the best moisture barrier of the three, but translucency rather than clarity. It is environmentally stress-cracking sensitive with surfactants unless high molecular weight grades are used.

HPSC EXAM TRAP: density ordering
Correct ascending order of density: PP (0.90) < LDPE and LLDPE (0.91 to 0.925) < HDPE (0.94 to 0.965) < EVOH (1.12 to 1.20) < PET (1.33 to 1.40) < PVC (1.35 to 1.45) < PVDC (1.6 to 1.7). Polyolefins float in water; PET, PVC and PVDC sink.

1.4.2 Polypropylene: PP, BOPP, CPP

Isotactic PP (all methyl groups on one side, Ziegler-Natta catalysed) crystallises to a stiff, glossy polymer with the lowest density and the highest melting point of the common polyolefins. Consequences:

- Retortable and hot-fillable (Tm 160 to 170 °C) whereas PE deforms at 121 °C.
- Poor low-temperature impact: homopolymer PP becomes brittle below about 0 °C, so frozen-food and retort grades use ethylene-propylene random or block copolymers.
- BOPP (biaxially oriented, tenter frame) gives high clarity, stiffness, twist retention and improved moisture barrier; metallised BOPP is the standard snack-food film; acrylic or PVDC coatings add sealability and barrier.
- CPP (cast polypropylene, chill-roll cast, unoriented) is soft, tough, clear and heat-sealable at 150 to 170 °C, and is the inner food-contact sealant layer of retort pouches because it survives 121 to 135 °C sterilisation without delaminating or losing seal strength. Random copolymer CPP (ethylene-modified) is preferred for low-temperature toughness.

1.4.3 Polyethylene terephthalate (PET)

PET is a condensation polyester of purified terephthalic acid (or dimethyl terephthalate) and ethylene glycol, usually polymerised with an antimony trioxide catalyst (hence the antimony specific migration limit of 0.04 mg/kg). Bottles are injection moulded as preforms, reheated to about 100 °C (above Tg) and stretch-blow moulded; biaxial stretching induces strain crystallisation, giving high strength, clarity and the gas barrier that holds 3 to 4 volumes of carbon dioxide in carbonated soft drinks for several months. Heat-set PET tolerates hot filling at 85 to 90 °C; crystallised CPET trays are oven and microwave stable.

Acetaldehyde and why a scavenger is needed: during melt processing at 270 to 290 °C PET undergoes thermal chain scission to vinyl ester end groups, which rearrange to release acetaldehyde (CH3CHO). Acetaldehyde dissolves in the bottle wall and slowly migrates into the contents, giving a fruity, sweetish off-flavour detectable in still water at about 10 to 20 µg/L (it is masked in flavoured colas but obvious in packaged drinking water). Control measures: low-acetaldehyde resin grades, solid-state polycondensation, careful melt temperature and residence time, and acetaldehyde scavengers such as anthranilamide (2-aminobenzamide), added at a few hundred mg/kg, which reacts with acetaldehyde to form a non-volatile quinazolinone. The Bureau of Indian Standards specification for PET bottles for packaged drinking water prescribes acetaldehyde control for this reason.

Barrier enhancement for oxygen-sensitive beverages (beer, juice): multilayer PET/EVOH/PET or PET/MXD6 nylon, plasma coatings of silicon oxide (SiOx) or diamond-like carbon on the inner surface, and oxygen-scavenging resins (cobalt-catalysed MXD6).

1.4.4 PVC and PVDC

PVC: polymer of vinyl chloride monomer (VCM is a confirmed human carcinogen; residual VCM in packaging is restricted to 1 mg/kg and non-detectable migration under EU rules). Unplasticised PVC (uPVC) is rigid and clear and was widely used for mineral water and edible oil bottles in India before PET displaced it; it has good oil and grease resistance, moderate oxygen barrier, and only moderate moisture barrier. Plasticised PVC (30 to 40 percent plasticiser such as DEHP, DOA, DINP) gives cling film with self-adhesion and high oxygen permeability (useful for fresh meat bloom), but plasticiser migration into fatty foods is the chief regulatory concern (Part 3). PVC softens at about 80 °C (not retortable) and liberates hydrogen chloride and dioxins on uncontrolled incineration.

PVDC: polyvinylidene chloride copolymerised with vinyl chloride or methyl acrylate (Saran, Dow, 1933). Its high chlorine content and high crystallinity give an outstanding barrier to oxygen, water vapour and aromas, and unlike EVOH its barrier does not deteriorate with humidity. Because pure PVDC is hard to process (it degrades near its melting point, releasing HCl), it is mostly used as a thin coating (K-coating) on cellophane, BOPP, PET and nylon, as a shrink film for cheese and processed meat, and as chub-pack film. Drawbacks: yellowing on ageing, chlorine content (environmental objection), not suitable for high-temperature retorting as a monolayer.

> HPSC EXAM TRAP: PVC is not PVDC
> PVC = one chlorine per repeat unit, moderate barrier, plasticiser migration issue. PVDC = two chlorines per repeat unit, excellent dual barrier, used as a coating. A statement "PVC offers excellent oxygen and moisture barrier" is false; the same sentence about PVDC is true.

1.4.5 Ethylene vinyl alcohol (EVOH)

EVOH is made by hydrolysing ethylene-vinyl acetate copolymer (pure polyvinyl alcohol cannot be melt-processed and is water-soluble, so ethylene is copolymerised in). The dense hydrogen bonding between hydroxyl groups and the high crystallinity give the lowest oxygen permeability of any melt-processable polymer (dry OTR of the order of 0.1 to 1 $cm^3/m^2 \cdot day$ for a 25 µm film, two to three orders of magnitude better than PET). Trade-offs:

- Lower ethylene content (27 to 32 mol percent): best gas barrier, worst moisture sensitivity, harder to process.
- Higher ethylene content (44 to 48 mol percent): better processability and humidity tolerance, poorer barrier.
- At relative humidity above about 75 percent, water plasticises the hydroxyl network and OTR rises ten to a hundred fold; after retorting the laminate suffers "retort shock" (temporary barrier loss) that recovers as the layer dries.

Therefore EVOH is never used alone: it is co-extruded as a thin core between polyolefin skins with maleic-anhydride-grafted tie layers (for example PP/tie/EVOH/tie/PP), where the hydrophobic skins keep the core dry. Applications: squeezable ketchup and mayonnaise bottles, retortable transparent trays and pouches (foil replacement for microwaveable products), cheese and processed meat films, modified atmosphere trays.

1.5 Retort pouches

Definition and origin

A retort pouch is a flexible, heat-sealable, hermetically sealed laminate container capable of withstanding thermal sterilisation at 115 to 135 °C (commonly 121 °C for 20 to 40 minutes, delivering commercial sterility expressed as an F0 value) so that low-acid foods are shelf stable at ambient temperature without refrigeration. It was developed in the 1960s and 1970s by the United States Army Natick Laboratories with

Reynolds Metals and Continental Flexible Packaging for military rations (Meal, Ready-to-Eat) and is now used for curries, ready meals, tuna, pet food and rice.

Laminate architecture

Layer (outside to inside)	Typical gauge	Material	Function
1. Outer	12 µm	Biaxially oriented PET (BOPET)	Printability (reverse printing protects ink from food and abrasion), tensile strength, scuff and heat resistance against the sealing jaws
2. Barrier	9 to 12 µm	Aluminium foil (soft annealed)	Essentially total barrier to oxygen, water vapour, light and aroma when laminated; gives the 18 to 36 month shelf life
3. Toughness (4-ply only)	15 µm	Biaxially oriented nylon 6 (OPA)	Puncture, flex-crack and abuse resistance; protects the foil from cracking in bony or sharp-edged products (meat, fish, ration packs)
4. Inner sealant	70 to 100 µm	Cast polypropylene (CPP), often ethylene-modified random copolymer	Food-contact layer; heat seals at about 160 °C, withstands 121 to 135 °C retorting, inert, low taint; thick enough to bridge seal-area contamination

The three-ply pouch is PET / Al / CPP; the four-ply pouch is PET / Al / OPA / CPP (nylon placed between the foil and the sealant). Layers are bonded by adhesive lamination with two-component polyurethane adhesives, which must be fully cured before filling to avoid primary aromatic amine migration (Part 3). Transparent and microwaveable retort pouches replace foil with PET/SiOx, PET/AlOx or EVOH-based barrier layers at some sacrifice of shelf life.

Heat penetration dynamics versus the cylindrical tinplate can

Thermal sterilisation of a conduction-heating food is unsteady-state conduction, and the time for the slowest-heating point (cold spot) to reach a target temperature scales with the square of the characteristic conduction distance. A 307 × 409 can (73 mm diameter, 115 mm height) has a cold-spot conduction path of about 36 mm; a filled retort pouch is a thin slab 15 to 25 mm thick, with a half-thickness of only 8 to 12 mm. Consequences:

1. Process time at the same retort temperature is cut by 30 to 50 percent for the same F0, because the whole pouch heats almost uniformly.
2. Less overcooking of the surface layers while waiting for the centre: better retention of thiamine, vitamin C, colour, texture and flavour; the product tastes closer to freshly cooked.
3. Lower energy per unit, faster cooling, lower steam and water consumption.
4. Requirements that cans do not impose: the pouch must be processed in an overpressure retort (steam-air or water immersion or spray) because internal vapour pressure at 121 °C would otherwise burst the seals; pouches are held flat in racks to control thickness; residual air must be minimised (under about 10 mL, by vacuum or steam-flush sealing) since trapped air insulates, floats the pouch and promotes oxidation; seal areas must be clean, and seals are tested for strength, burst pressure and leakage.

5. After processing, pouches require a carton for distribution (less rigid than cans) and quality control includes an incubation test (typically 37 °C for 10 days, with 55 °C incubation for thermophilic checks) as prescribed in FSSAI's commercial sterility requirements for canned and retorted foods.

Advantages and limitations relative to tinplate cans

Aspect	Retort pouch	Tinplate can
Heat penetration	Fast (thin slab)	Slow (large cylinder); surface overcooks
Nutrient and sensory quality	Higher retention	Lower
Weight and storage	Light, flat when empty, 85 percent less storage space	Heavy, bulky
Consumer convenience	Tear-notch opening, boil-in-bag reheating, microwaveable if foil-free	Needs opener; cannot be microwaved
Shelf life	18 to 36 months (foil type)	24 to 60 months
Line speed and cost	Slower filling, pouch cost higher per unit, seal integrity more critical	High-speed seaming, well-established inspection
Physical protection	Needs carton; puncture risk	Rigid, stackable
Metal pickup	None (polymer contact)	Tin and iron pickup in plain cans; lacquer migration issues in coated cans
Recycling	Multi-material laminate, difficult	Readily recycled magnetically

Part 2. Active, Intelligent & Atmosphere Packaging

PART 2: ACTIVE, INTELLIGENT AND MODIFIED ATMOSPHERE PACKAGING

2.0 Definitions that examiners quote

- Active packaging: packaging in which the package, the product and the environment interact deliberately to extend shelf life, improve safety or sensory quality, while maintaining quality; the EU definition (Regulation (EC) 1935/2004 and Regulation (EC) 450/2009) is that active materials "deliberately incorporate components that would release or absorb substances into or from the packaged food or the environment surrounding the food". Active packaging acts.
- Intelligent (smart) packaging: materials and articles that monitor the condition of packaged food or the environment surrounding the food and communicate that information. Intelligent packaging informs; it does not change the food.
- Modified atmosphere packaging (MAP): replacement of the air in the package with a defined gas mixture at the time of packing, after which the atmosphere is not controlled (distinguish from controlled atmosphere storage, where gas composition is actively maintained in a store).

Feature	Active packaging	Intelligent packaging
Purpose	Change the internal environment or the food to extend shelf life	Sense and report the state of the food, the package or its history
Interaction	Absorbs or releases substances (oxygen, carbon dioxide, ethylene, moisture, antimicrobials, flavours)	No substance exchange intended; optical, chemical or electronic signal
Examples	Oxygen scavenger sachet, ethylene absorber, antimicrobial film, desiccant, carbon dioxide emitter, ethanol emitter	Time-temperature indicator, freshness indicator, leak (integrity) indicator, RFID tag, barcode, biosensor
Regulatory issue	Migration of the active substance into food (must be permitted substances; sachets must not be ingested)	Indicator chemicals must not migrate; label information must not mislead

2.1 Oxygen scavengers

Oxygen drives lipid oxidation (rancidity, loss of essential fatty acids and vitamins A, C, E), oxidative browning and pigment loss, and growth of aerobic spoilage organisms including moulds. Vacuum and gas flushing leave 0.5 to 2 percent residual oxygen; a scavenger drives headspace oxygen below 0.01 percent (100 ppm) and keeps absorbing oxygen that permeates through the film during storage.

Iron-based sachets (Ageless by Mitsubishi Gas Chemical, 1977; FreshPax, Oxy-Guard)

Finely divided reduced iron powder with an electrolyte (sodium chloride) and a moisture donor, packed in an oxygen-permeable but food-grade sachet. The reaction is electrochemical rusting:

$4Fe + 3O2 + 6H2O \rightarrow 4Fe(OH)3$

Stoichiometry: 4 moles of iron (about 223 g) consume 3 moles of oxygen (about 67 L at standard conditions), so 1 g of iron absorbs roughly 300 mL of oxygen; sachets are graded by capacity (for example 100 or 200 mL oxygen). Moisture-dependent types need a food water activity above about 0.65; self-reacting types carry their own moisture and work in dry foods. Advantages: cheap, high capacity, rapid.

Disadvantages: sachet can be accidentally ingested (must be labelled "do not eat"; not permitted loose in some infant foods), iron is detected by in-line metal detectors so sachet insertion must be sequenced after detection or non-ferrous scavengers used, sachets are unsuitable for liquids and for microwave heating, and the pack can collapse as oxygen is removed (compensated by combining with a carbon dioxide emitter).

Non-iron (non-metallic) systems

System	Chemistry	Form and use
Ascorbic acid or ascorbate (often copper or iron catalysed)	Oxidised to dehydroascorbic acid	Sachets and closures; metal-detector compatible
Sulfite (sodium sulfite with catalyst)	Oxidised to sulfate	Sachets
Catechol, gallic acid, hydroquinone	Phenolic oxidation	Sachets, films
Enzymatic: glucose oxidase with catalase	Glucose + O_2 → gluconic acid + H_2O_2; catalase removes peroxide	Films and closures; needs moisture; also used in liquids
Photosensitive dye plus singlet-oxygen acceptor	Light-activated oxidation of an unsaturated compound	Transparent films, activated on the packing line
Unsaturated polymer (polybutadiene) with cobalt catalyst	Autoxidation of the polymer	Film layer
Oxygen-scavenging nylon MXD6 with cobalt salt (Oxbar, Amosorb)	Polymer oxidation	Integral layer in PET beer and juice bottles
Palladium catalyst with hydrogen in headspace	$2H_2 + O_2 \rightarrow 2H_2O$	Crown liners (beer)

Typical applications: bakery (mould-free life of chapati, bread and cakes extended several times), nuts and snacks, dried meat and fish, coffee, cheese, pharmaceuticals, and combined with modified atmosphere for cooked meats.

2.2 Moisture regulators and desiccants

Water vapour causes caking of powders, loss of crispness, mould growth, drip and clouding. Controls:

Agent	Mechanism	Properties
Silica gel (amorphous SiO_2)	Physical adsorption in micropores	Absorbs 30 to 40 percent of its own weight; regenerable; inert; indicator grade formerly used cobalt chloride (blue when dry, pink when wet), now replaced by non-toxic indicators because cobalt(II) chloride is classified as a carcinogen
Molecular sieves (synthetic zeolites 3A, 4A)	Size-selective adsorption in crystalline aluminosilicate cages	Reach very low humidity and work at elevated temperature; higher cost; used for sensitive pharmaceuticals and some dried foods
Clay (montmorillonite, bentonite)	Adsorption in layered silicate	Cheapest; effective at ambient humidity; natural; used in bulk sachets
Calcium oxide (quicklime)	Chemical reaction to calcium hydroxide, irreversible	High capacity at low humidity; alkaline, so used only in sachets (nori seaweed, dried fish)

Agent	Mechanism	Properties
Calcium chloride	Deliquescent; dissolves in absorbed water	Container desiccants during shipping
Superabsorbent polymer pads (polyacrylate)	Absorbs liquid drip	Pads under fresh meat, poultry and fish trays; also absorbs melted ice in seafood
Two-way humidity regulators (Humidipak)	Saturated salt solution behind a membrane releases or absorbs vapour	Maintains a fixed RH for products such as dried fruit and tobacco
Anti-fog films	Surfactant migrates to film surface to spread condensation	Visibility of chilled produce and salads

Related emitters: ethanol-vapour sachets (Ethicap, ethanol adsorbed on silica) extend mould-free life of bakery products; flavour and aroma releasing films are also classified as active.

2.3 Ethylene absorbers

Ethylene (C_2H_4) is the gaseous ripening hormone; concentrations as low as 0.1 ppm trigger the climacteric rise in respiration of bananas, mangoes, apples, tomatoes and avocados, accelerate chlorophyll loss in leafy vegetables and cause russet spotting of lettuce. Removing it from the package or store extends shelf life.

Potassium permanganate system: $KMnO_4$ (4 to 6 percent) impregnated on a porous inert carrier (alumina, silica gel, zeolite, vermiculite, perlite) oxidises ethylene stepwise to acetaldehyde, then acetic acid, then carbon dioxide and water; the overall reaction can be written $3C_2H_4 + 12KMnO_4 \rightarrow 12MnO_2 + 12KOH + 6CO_2 + 6H_2O$. The colour change from purple (permanganate) to brown (manganese dioxide) indicates exhaustion. Because permanganate is toxic and strongly oxidising it is never incorporated in food-contact film; it is used only in sachets, blankets or filters in cartons and cold stores. Other absorbers: activated carbon with palladium catalyst, zeolite- or clay-loaded polyethylene films (sold in Japan and Korea, limited capacity), and ethylene-oxidising catalysts in ripening rooms. Note that 1-methylcyclopropene (SmartFresh) is a ripening inhibitor applied as a gas treatment, not a packaging absorber.

2.4 Antimicrobial packaging

Antimicrobial agents are either released from the package to the food surface (migrating systems, suitable where surface contamination governs spoilage, such as sliced meat and cheese) or immobilised on the film surface (non-migrating, contact-active systems).

Agent	Nature	Target and notes
Nisin	Bacteriocin from Lactococcus lactis (INS 234, permitted additive in India for cheese and some other foods)	Gram-positive bacteria including Listeria monocytogenes and Clostridium spores (with heat); inactive on Gram-negatives unless combined with chelators
Natamycin (pimaricin)	Polyene antifungal	Mould and yeast on cheese rinds and sausages
Lysozyme	Enzyme hydrolysing peptidoglycan	Gram-positive bacteria; used in cheese
Organic acids and salts (sorbate, benzoate, propionate, lactate)	Weak acid preservatives in edible coatings or films	Yeasts, moulds, bacteria; pH dependent

Agent	Nature	Target and notes
Essential oils (carvacrol, thymol from oregano and thyme; cinnamaldehyde; eugenol; allyl isothiocyanate from mustard)	Volatile phenolics and isothiocyanates disrupting membranes	Broad spectrum; vapour-phase action in headspace; strong odour limits dose; GRAS status helps approval
Chitosan	Cationic polysaccharide from chitin	Film-forming antimicrobial coating for fruit
Silver (ions, nanoparticles, silver zeolite, silver-substituted glass)	Ag^+ binds thiol groups of enzymes, damages membranes and DNA	Broad spectrum, long-lasting; approved in Japan and the US; EU requires specific migration limit for silver of 0.05 mg/kg; nanoparticle migration and resistance are open regulatory questions
Zinc oxide and titanium dioxide nanoparticles	Photocatalytic reactive oxygen species	Films and coatings; titanium dioxide status under review in the EU
Chlorine dioxide and sulfur dioxide releasers	Gas generation from sachet or pad (sodium metabisulfite pads on table grapes)	Botrytis control in grapes during export
Triclosan	Synthetic phenol	Restricted in food contact in the EU; exam reference only

2.5 Carbon dioxide emitters and absorbers

Carbon dioxide at 20 to 100 percent is bacteriostatic and fungistatic (it dissolves in the aqueous phase, lowers pH and interferes with enzymes of Pseudomonas, Enterobacteriaceae and moulds), but it dissolves into the food, which causes pack collapse in modified atmosphere packs. A CO_2 emitter compensates:

- Sodium bicarbonate with citric or ascorbic acid, activated by product moisture: $NaHCO_3 + H^+ \rightarrow Na^+ + CO_2 + H_2O$. Used in pads under fresh meat, poultry and fish.
- Dual-function sachets (iron oxygen scavenger with ferrous carbonate or ascorbate plus bicarbonate, for example Ageless G) absorb oxygen and emit an equal volume of carbon dioxide to keep the pack full.

Carbon dioxide absorbers: freshly roasted coffee releases carbon dioxide for days and would burst a hermetic pack; calcium hydroxide sachets absorb it, $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$; the alternative is a one-way degassing valve in the pack. Absorbers are also used for kimchi and other fermenting foods.

2.6 Modified atmosphere packaging and vacuum packaging

Gases used: oxygen (maintains the cherry-red oxymyoglobin of fresh meat, allows respiration of produce, prevents anaerobic growth), carbon dioxide (antimicrobial, soluble), nitrogen (inert filler preventing collapse, displaces oxygen in snacks), and in some countries carbon monoxide (colour fixation of meat; not permitted in India and the EU) and argon.

Product	Typical gas mix	Rationale
Fresh red meat	70 to 80 percent O_2, 20 to 30 percent CO_2	High oxygen keeps bloom; CO_2 suppresses Pseudomonas
Cured meat, poultry, sliced cooked meats	20 to 35 percent CO_2, remainder N_2	No oxygen to avoid rancidity and colour fading
White fish	40 percent CO_2, 30 percent O_2, 30 percent N_2	Oxygen included to inhibit Clostridium botulinum in low-oxygen chilled fish
Oily fish	40 to 60 percent CO_2, remainder N_2	No oxygen to avoid rancidity

Product	Typical gas mix	Rationale
Bakery products	50 to 100 percent CO2	Mould inhibition
Hard cheese	100 percent CO2 or CO2 and N2	Mould inhibition
Snacks, nuts, coffee	100 percent N2	Prevents oxidation and crushing
Fresh-cut produce (equilibrium MAP)	3 to 5 percent O2, 3 to 10 percent CO2	Respiring product; film permeability or microperforation sets the equilibrium; too little oxygen causes fermentation

Safety points for the Food Safety Officer: MAP does not sterilise; it only slows spoilage at chill temperature, and by suppressing spoilage organisms it may let pathogens such as Listeria monocytogenes and non-proteolytic Clostridium botulinum reach hazardous levels without obvious spoilage, so strict temperature control (below 3.3 °C for vacuum-packed fish) and shelf-life validation are essential. Vacuum packaging removes air without replacing it and depends on an oxygen-barrier film (PA/PE, PVDC-coated) and refrigeration.

2.7 Intelligent packaging systems

2.7.1 Time-temperature indicators (TTIs)

A TTI is a device whose irreversible, visible response integrates the temperature history (full-history TTI) or signals only that a threshold was crossed (critical temperature indicator, partial history). The reaction rate follows Arrhenius kinetics; a TTI is matched to a food when its activation energy is within about 25 kJ/mol of the activation energy of the food's deterioration (for example microbial growth in chilled fish, about 60 to 90 kJ/mol).

Type	Commercial example	Mechanism	Reading
Diffusion based	3M MonitorMark	A fatty acid ester with a selected melting point melts on temperature abuse and a dyed front migrates along a porous wick	Distance travelled indicates time above threshold
Polymerisation based	Lifelines Fresh-Check and Fresh-Scan	Solid-state polymerisation of a diacetylenic monomer; the polymer darkens at a rate dependent on temperature	Centre of the bullseye darker than the reference ring means end of shelf life; the label must be stored frozen before use
Enzymatic	VITSAB Check Point (Sweden)	Lipase hydrolyses a triglyceride substrate; released fatty acids lower pH and a mixed indicator shifts from green through yellow	Activated at packing by breaking the barrier between enzyme and substrate
Microbiological	TopCryo (Cryolog, France)	Lactic acid bacteria in a gel grow and acidify; indicator colour shifts from green to red	Mimics microbial growth in the food
Photochemical	OnVu (Ciba and Freshpoint)	Benzylpyridine photochromic ink charged with ultraviolet light fades at a temperature-dependent rate	Fading beyond a reference means abuse

Type	Commercial example	Mechanism	Reading
Nanoparticle	Gold nanoparticle aggregation systems	Colour change on aggregation	Experimental

Uses: cold-chain monitoring of vaccines, chilled and frozen foods, milk, seafood exports; supports first-expired-first-out instead of first-in-first-out logistics.

2.7.2 Freshness and spoilage indicators

These respond to metabolites of spoilage rather than to temperature history.

- Volatile amines: spoilage of fish and meat releases trimethylamine (from trimethylamine oxide by Shewanella and Photobacterium), dimethylamine, ammonia and biogenic amines, collectively measured as total volatile basic nitrogen (TVB-N). Amines are alkaline and raise headspace pH; indicator films containing bromocresol green, bromophenol blue, bromothymol blue, methyl red, chlorophenol red or natural anthocyanins change colour (COX Technologies FreshTag for fish; many anthocyanin-based research films).
- Hydrogen sulfide: poultry spoilage indicators using myoglobin-based pigments that convert to sulfmyoglobin (green).
- Carbon dioxide: bromothymol-blue or methyl-red based indicators for kimchi and MAP integrity.
- Aroma: RipeSense (New Zealand) senses aromatic volatiles from ripening pears and changes from red to orange to yellow.
- Pathogen or toxin biosensors: antibody-based labels (for example FoodSentinel barcode that becomes unreadable when a target pathogen binds) remain largely developmental.

2.7.3 Integrity (leak) indicators: oxygen indicators based on redox dyes

The classic oxygen indicator tablet (Ageless Eye, Mitsubishi) contains methylene blue, a reducing sugar (glucose) and an alkali in a gel. In an oxygen-free pack (below about 0.1 percent oxygen) the dye is in its reduced leuco form and appears pink; when oxygen enters through a pinhole or a failed seal (above about 0.5 percent) methylene blue is re-oxidised and turns blue, usually within minutes. It is used inside packs with oxygen scavengers and in modified atmosphere packs to show seal failure, and it reverts if oxygen is scavenged again, which is a limitation for tamper evidence. Related systems use ultraviolet-activated titanium dioxide and methylene blue in printed inks for irreversible indication.

2.7.4 Data carriers: barcodes and RFID

- Barcodes: linear EAN-13 (the 13-digit code on retail packs, administered in India by GS1 India, prefix 890) carry only an identifier; two-dimensional GS1 DataMatrix and QR codes carry batch, expiry and serial numbers and support consumer access to FSSAI licence details and recall information.
- RFID (radio-frequency identification): a tag (microchip plus antenna) is read without line of sight; passive tags are powered by the reader field, active tags carry a battery. Frequencies: low frequency about 125 kHz (short range, animal identification), high frequency 13.56 MHz (near-field communication, smart phones, library and item level), ultra-high frequency 860 to 960 MHz (pallet and case level, several metres read range). Benefits: automatic traceability (one step back, one step forward as required by FSSAI licensing conditions and Codex), cold-chain data when combined with temperature sensors, rapid recall, anti-counterfeiting. Limitations: cost per tag, interference from metals and liquids, data privacy.

HPSC EXAM TRAP: active versus intelligent
A time-temperature indicator, a freshness indicator and an oxygen leak indicator are intelligent packaging (they inform). An oxygen scavenger, an ethylene absorber and an antimicrobial film are active packaging (they act). Statements that classify a scavenger as intelligent, or an indicator as active,

are wrong. Also, a TTI tracks temperature history, not the actual microbial state of the food; a freshness indicator tracks spoilage metabolites, not temperature.

Part 3. Migration Chemistry & FSSAI Packaging Regulations 2018

PART 3: MIGRATION CHEMISTRY AND FSS (PACKAGING) REGULATIONS, 2018

3.1 Migration from first principles

Migration is the mass transfer of substances from a food contact material into the food (or a simulant) by diffusion. The migrating substances are starting substances (monomers, catalysts such as antimony), additives (plasticisers, antioxidants, slip agents, colourants), processing aids, adhesive and ink components, and non-intentionally added substances (NIAS) such as oligomers, degradation products and impurities.

Kinetics: migration obeys Fick's second law; early migration grows with the square root of contact time and later approaches equilibrium set by the partition coefficient between polymer and food. Migration increases with temperature (retorting, microwaving, hot filling), contact time, fat content of the food (lipophilic migrants partition into oil), decreasing molecular weight of the migrant (substances above about 1000 Da are considered not to be absorbed from the gut and are exempt from toxicological evaluation), and polymer mobility (polyolefins have high diffusivity, PET and polystyrene low). Migration also occurs through the gas phase (volatile ink components and mineral oil from paperboard) and by set-off (printed outer surface transferring ink to the food-contact side in a reel or stack).

Functional barrier concept: a layer (for example aluminium foil or thick PET) that reduces migration from layers behind it to below the detection limit allows the use of non-approved substances behind it, subject to a no-carcinogen rule.

3.2 Overall migration versus specific migration

Criterion	Overall migration (OM)	Specific migration (SM)
What is measured	Total mass of all non-volatile substances transferred from the material into a food simulant, irrespective of identity	Amount of a single named substance (monomer, additive, metal) transferred
Basis of the limit	Inertness of the material (a general quality criterion), not toxicology	Toxicology: the specific migration limit (SML) is derived from the tolerable daily intake
How measured	Gravimetric: evaporate the simulant to dryness and weigh the residue (or extract from oil by solvent and weigh)	Instrumental: HPLC, GC-MS, LC-MS/MS, AAS or ICP-MS for the specific analyte
Units	mg/dm² of contact surface or mg/kg of food or simulant	mg/kg of food or simulant (occasionally mg/dm²)
Statutory ceiling	60 mg/kg or 10 mg/dm² (FSSAI 2018 Schedule IV; EU Regulation 10/2011)	Substance specific, for example DEHP 1.5 mg/kg, antimony 0.04 mg/kg, zinc 25 mg/kg
Abbreviation	OML (overall migration limit)	SML (specific migration limit); also QM (maximum residual quantity in the material) where migration cannot be measured

The arithmetic link examiners love: regulators assume a 60 kg adult eats 1 kg of food per day that has been in contact with 6 dm^2 of packaging (a cube of 1 litre volume has 6 dm^2 of surface). Therefore 10 mg/dm^2 × 6 dm^2/kg = 60 mg/kg. Both figures express the same ceiling; for small containers (below 500 mL) and sheet materials the surface-based figure is used, for larger containers the mass-based figure.

HPSC EXAM TRAP: overall migration is not a toxicological limit
Passing the overall migration test does not prove safety; a material can be below 60 mg/kg overall and still exceed the SML of a potent migrant such as BPA or a primary aromatic amine. Conversely a high overall migration (for example from a plasticised PVC) can occur with every individual SML met. Both tests are required.

3.3 Food simulants

Foods are too variable for direct testing, so standardised liquids or solids that mimic the extractive power of food classes are used.

Simulant (EU Regulation 10/2011, Annex III)	Composition	Represents
Simulant A	10 percent ethanol (v/v) in water	Hydrophilic foods, aqueous foods with pH above 4.5
Simulant B	3 percent acetic acid (w/v) in water	Acidic aqueous foods with pH below 4.5 (fruit juices, pickles, soft drinks, tomato products)
Simulant C	20 percent ethanol (v/v)	Alcoholic foods up to 20 percent alcohol and foods with a lipophilic character (relevant organic ingredients)
Simulant D1	50 percent ethanol (v/v)	Alcoholic foods above 20 percent alcohol, oil-in-water emulsions, milk and dairy products
Simulant D2	Vegetable oil (refined olive or sunflower oil with less than 1 percent free fatty acids)	Fatty foods with free fat on the surface; substitutes of 95 percent ethanol and iso-octane are permitted when oil analysis is not feasible
Simulant E	Poly(2,6-diphenyl-p-phenylene oxide), trade name Tenax, modified polyphenylene oxide (MPPO), particle size 60 to 80 mesh	Dry foods (cereals, biscuits, powders, rice); captures volatile and semi-volatile migrants from paper and board and plastics

Standard test conditions (EU OM1 to OM7, and the same logic in IS 9845): 10 days at 40 °C for long-term storage at room temperature; 10 days at 20 °C for frozen or refrigerated storage; 2 hours at 70 °C for hot fill; 1 hour at 100 °C for boiling; 30 minutes at 121 °C for retort sterilisation; 2 hours at 175 °C for ovenable trays in Tenax. Indian practice under IS 9845 (Determination of overall migration of constituents of plastics materials and articles intended to come in contact with foodstuffs) uses aqueous, acidic, alcoholic and fatty simulants (distilled water, 3 percent acetic acid, aqueous ethanol and n-heptane) with time-temperature conditions matched to the intended use; the FSSAI 2018 Regulations refer to IS 9845 for overall migration testing of plastics.

HPSC EXAM TRAP: simulant letters
Memorise the ethanol ladder: A = 10 percent, C = 20 percent, D1 = 50 percent. B is the only acid (3

percent acetic acid). D2 is the only oil. E is the only solid (Tenax). A question stating "Simulant B is 10 percent ethanol" or "Simulant E is vegetable oil" is wrong.

3.4 Migrants of concern

Migrant	Source in packaging	Health concern	Regulatory reference points
Bisphenol A (BPA)	Epoxy-phenolic can coatings, polycarbonate bottles and water dispensers, lid linings, thermal receipt paper	Endocrine disruptor (oestrogenic, anti-androgenic), effects on reproduction, metabolism and immune system; classified by EFSA in 2023 with a tolerable daily intake lowered to 0.2 nanograms per kg body weight per day	EU SML reduced from 0.6 to 0.05 mg/kg (2018); EU ban on BPA in polycarbonate infant feeding bottles (2011) and Regulation (EU) 2024/3190 prohibiting BPA in food contact materials with transition periods from 2025; Indian Standard IS 14625 for plastic feeding bottles revised to exclude polycarbonate; FSSAI monitors through its packaging and infant food provisions
Phthalates (DEHP, DBP, BBP, DINP, DIDP)	Plasticisers (up to 40 percent) in flexible PVC cling film, tubing, gloves, conveyor belts, and PVC plastisol gaskets in metal lids of glass jars (migration into oily foods such as pesto and sauces was a major European finding)	Reproductive and developmental toxicity (anti-androgenic), DEHP and DBP classified reprotoxic category 1B; liver effects	FSSAI Schedule IV SML for DEHP 1.5 mg/kg; EU SMLs: DEHP 1.5, DBP 0.3, BBP 30, DINP plus DIDP 9 mg/kg with group limits; DEHP use restricted to repeated-use, non-fatty applications
Primary aromatic amines (PAAs: aniline, 2,4-toluenediamine, 4,4'-methylenedianiline)	Two-component polyurethane laminating adhesives based on aromatic isocyanates (TDI, MDI): unreacted isocyanate migrates and hydrolyses to the amine; also from certain azo dyes and pigments and from black nylon kitchen utensils	Several are genotoxic carcinogens (bladder cancer); aniline is a suspected carcinogen	EU Regulation 10/2011: no detectable migration, detection limit 0.01 mg/kg for the sum and 0.002 mg/kg for individually classified carcinogenic PAAs (amendment 2020); mitigation: full adhesive cure (several days) before filling, aliphatic isocyanates, solvent-free adhesives
Benzophenone and related photoinitiators (4-methylbenzophenone, isopropylthioxanthone ITX)	Ultraviolet-curable printing inks and varnishes on cartons; migrate through paperboard by gas phase and by set-off	Liver and kidney toxicity in animals; some are classified as possible carcinogens	EU SML for benzophenone 0.6 mg/kg; the 2005 Nestle infant milk recall in Italy (ITX set-off from Tetra Pak printing) and the 2009 4-methylbenzophenone in breakfast cereals incident drove ink regulation; Indian Standard IS 15495 for printing inks and the Swiss Ordinance positive list are the references; FSSAI requires the printed surface not to touch food

Migrant	Source in packaging	Health concern	Regulatory reference points
Mineral oil hydrocarbons (MOSH, MOAH)	Recycled paper and board containing newspaper offset inks; jute bag batching oil; lubricants	MOSH accumulate in liver and lymph nodes; MOAH include genotoxic aromatics	Scientific basis for the FSSAI prohibition of newspaper wrapping and for the European 2 mg/kg MOAH action threshold; FSSAI advisory on jute bags for food grains to use vegetable-oil batching
Styrene	Polystyrene cups and trays	Possible carcinogen (IARC 2A), neurotoxic at high exposure	Migration limited by specifications on residual monomer
Vinyl chloride monomer	PVC	Confirmed human carcinogen (angiosarcoma of liver)	Residual monomer in material 1 mg/kg, non-detectable migration
Acetaldehyde	PET bottles	Off-flavour; possibly carcinogenic by inhalation	SML 6 mg/kg (EU); organoleptic control in PET water bottles
Antimony	PET polymerisation catalyst	Toxic at high intake	SML 0.04 mg/kg (FSSAI Schedule IV and EU)
Melamine and formaldehyde	Melamine-formaldehyde tableware under hot acidic conditions and microwaving	Kidney stones and renal failure (melamine); formaldehyde carcinogen	EU SML melamine 2.5 mg/kg, formaldehyde 15 mg/kg; melamine as adulterant limit in foods: 1 mg/kg in powdered infant formula and 2.5 mg/kg in other foods (Codex, adopted by FSSAI)
Heavy metals (lead, cadmium)	Pigments and stabilisers in plastics, ceramic glazes, enamel, solder	Neurotoxic (lead), nephrotoxic (cadmium)	IS 9833 purity requirements for colourants in plastics; lead solder banned in cans; lead and cadmium release limits for ceramic ware
Per- and polyfluoroalkyl substances (PFAS)	Grease-proof paper, microwave popcorn bags	Persistent, immune and liver effects	Restrictions in Denmark and the EU; emerging issue

3.5 Food Safety and Standards (Packaging) Regulations, 2018

Legislative lineage

1. Food Safety and Standards Act, 2006: Section 3(1)(zz) defines unsafe food and includes food rendered injurious by the packaging in which it is contained; Section 16 empowers the Food Authority to specify standards for packaging; packaging that renders food unsafe attracts penalties under Section 59 (unsafe food) and related sections.
2. FSS (Packaging and Labelling) Regulations, 2011: combined packaging and labelling provisions.
3. FSS (Packaging) Regulations, 2018: notified in the Gazette on 24 December 2018 and enforced from 1 July 2019, carving packaging out as a stand-alone regulation; the labelling provisions were later replaced by the FSS (Labelling and Display) Regulations, 2020.
4. Interlocking rules: Plastic Waste Management Rules, 2016 (Ministry of Environment), which prohibit the use of recycled plastics and carry bags made of recycled plastic for storing, carrying, dispensing or packaging ready-to-eat or ready-to-drink food, set carry-bag thickness (raised to 75 µm from 30 September 2021 and 120 µm from 31 December 2022) and banned identified single-use plastic items

(plastic cutlery, straws, stirrers, earbuds, wrapping films around sweet boxes) from 1 July 2022; and Bureau of Indian Standards specifications referenced in the Schedules.

Structure of the 2018 Regulations

Provision	Content
Regulation 2: Definitions	Food packaging, primary packaging (in direct contact with food), secondary packaging, food contact materials, migration, overall migration, specific migration, food simulant, recycled plastics and related terms
Regulation 3: General requirements	Material must be suitable for the type of product and conditions of use; must preserve the food and not alter its odour, colour or taste; must not transfer constituents to food in quantities that endanger human health, cause unacceptable changes in composition or deteriorate organoleptic characteristics; must conform to the Indian Standards listed in the Schedules; colourants and pigments in plastics must conform to IS 9833 (list of pigments and colourants for use in plastics in contact with foodstuffs, pharmaceuticals and drinking water); printing inks must conform to IS 15495 (printing ink for food packaging), and the printed surface of packaging shall not come in direct contact with food; packaging must be free from pinholes, cracks and defects; newspaper and similar materials shall not be used for packing or wrapping food
Regulation 4: Specific requirements	Material-wise requirements for paper and board, glass, metal and metal alloys, and plastics, cross-referring to the Schedules; for plastics: conformity to listed IS specifications, overall migration not exceeding 60 mg/kg or 10 mg/dm² when tested per IS 9845, specific migration limits for listed metals and DEHP, and the prohibition on recycled plastics and on products made of recycled plastics for packaging, storing, carrying or dispensing food
Regulation 5: Conformity	The food business operator must obtain a certificate of conformity from a laboratory accredited by NABL for the packaging material used (migration testing), and must ensure multilayer or novel materials are evaluated for safety
Schedule I	Paper and paperboard materials: Indian Standards for kraft paper, greaseproof paper, paperboard, corrugated board, paper for food wrapping, with purity requirements (heavy metals, pentachlorophenol, formaldehyde, optical brighteners)
Schedule II	Glass containers: listed Indian Standards; freedom from defects
Schedule III	Metals and alloys: tinplate (IS 1993), tin-free steel, aluminium foil and containers, with lacquer requirements
Schedule IV	Plastic materials: positive list of Indian Standards (for example IS 10146 polyethylene, IS 10910 polypropylene, IS 12229 PET, IS 10151 PVC, IS 10142 polystyrene, IS 11704 ethylene vinyl acetate, IS 12247 nylon, IS 13601 polycarbonate, IS 11434 ionomers, IS 13576 polyacrylonitrile) and the migration limits tables

Statutory migration limits (Schedule IV, plastics)

Parameter	Limit
Overall migration limit	60 mg/kg of food or simulant, or 10 mg/dm² of contact surface, tested per IS 9845
Barium	1.0 mg/kg
Cobalt	0.05 mg/kg
Copper	5.0 mg/kg
Iron	48 mg/kg
Lithium	0.6 mg/kg
Manganese	0.6 mg/kg
Zinc	25 mg/kg
Antimony	0.04 mg/kg
Phthalic acid, bis(2-ethylhexyl) ester (DEHP)	1.5 mg/kg

Heavy metals of toxicological concern (lead, cadmium, mercury, hexavalent chromium) are controlled at source: colourants and pigments must meet IS 9833 purity limits, lead solder is prohibited, and the FSS (Contaminants, Toxins and Residues) Regulations, 2011 set limits in the food itself (for example lead in milk 0.02 mg/kg, tin in canned foods 250 mg/kg, mercury in fish 0.5 mg/kg and 1.0 mg/kg in predatory species). The EU Packaging and Packaging Waste Directive 94/62/EC sets a benchmark of 100 mg/kg for the sum of lead, cadmium, mercury and hexavalent chromium in packaging material; migration of lead from plastics under the EU specific limits is at the 0.01 to 1 mg/kg level depending on the article class. Candidates should quote the Gazette text of Schedule IV in the descriptive paper and avoid inventing figures.

Absolute prohibitions and their scientific basis

1. Newspaper and similar printed materials shall not be used for wrapping, packing or serving food (FSSAI advisory of 2016 incorporated in Regulation 3). Basis: newspaper inks contain mineral oil hydrocarbons, solvents, pigments with lead and other metals and sometimes chemical driers; the paper is handled unhygienically and carries microbial load; lipophilic ink components migrate readily into fried and oily street foods. The common sight of pakoras drained on newspaper is therefore a punishable violation.
2. Recycled plastics and products made of recycled plastics (including carry bags) shall not be used for packaging, storing, carrying or dispensing food. Basis: unknown prior use, contamination with non-food chemicals, degradation products and unapproved additives. Exception: FSSAI's subsequent amendment framework and Bureau of Indian Standards IS 17899 T (2022) allow recycled PET and other recycled plastics for food contact only when produced by an FSSAI-approved or certified food-grade recycling process with decontamination challenge testing, in line with Plastic Waste Management (Amendment) Rules; recycled material used behind a functional barrier is treated separately. In an examination answer: "prohibited, except certified food-grade recycled plastic produced under FSSAI-recognised processes".
3. Printed surface of packaging shall not come in direct contact with food, so printed recycled paper, printed carry bags and reverse-printed films where the print faces inward are not acceptable; printing inks must conform to IS 15495.
4. Packaging must not impart undesirable odour, colour or taste, and must not transfer substances harmful to health; this is the general safety clause under which any novel migrant can be acted upon.
5. Plastics in contact with food must be made only of listed polymers conforming to listed Indian Standards; carbon black and pigments only per IS 9833.

6. Tin coating and lacquers for metal containers must conform to the listed standards; lead-soldered cans are not permitted.

HPSC EXAM TRAP: absolute-qualifier phrasing on prohibitions
"Newspaper may be used for wrapping dry foods if clean": false, the prohibition is absolute. "Recycled plastics are permitted for food contact": false as a general statement, but "recycled plastics are permitted only when certified food-grade under FSSAI-recognised processes" is true. "The overall migration limit applies to glass": false, it applies to plastics (glass is treated as inert and controlled by specification). "The 2018 Packaging Regulations also cover labelling": false, labelling is in the 2020 Labelling and Display Regulations.

Migration testing workflow a Food Safety Officer should be able to describe

1. Identify the packaging material, layer structure and intended use (food type, contact time, temperature).
2. Select simulants from the food classification (aqueous, acidic, alcoholic, fatty, dry) and the time-temperature condition matching the worst foreseeable use.
3. Expose a known surface area (for example 1 dm^2) to a known volume of simulant (for example 100 mL, giving 6 dm^2 per kg convention) by total immersion, pouch filling or cell method.
4. Overall migration: evaporate the aqueous simulant to dryness, weigh the residue, correct for blank, express as mg/dm^2 and convert to mg/kg; for oil, determine the weight loss of the specimen corrected for oil uptake.
5. Specific migration: analyse the simulant by HPLC-UV or LC-MS/MS (phthalates, BPA, PAAs, benzophenone), GC-MS (volatiles, styrene, vinyl chloride), ICP-MS or AAS (metals).
6. Compare against the OML and SMLs, issue the NABL certificate of conformity, and in enforcement draw samples under Section 38 of the Act for referral to the Food Analyst.

Part 4A. Instrumental Food Analysis

PART 4A: INSTRUMENTAL FOOD ANALYSIS

4.0 Chromatography from first principles

Chromatography separates components of a mixture by their differential partitioning between a stationary phase and a mobile phase that moves past it. Key quantities:

- Retention factor $k = (tR - t0)/t0$, where tR is the retention time of the analyte and t0 the hold-up time of an unretained compound.
- Selectivity $\alpha = k2/k1$ for two adjacent peaks.
- Column efficiency as number of theoretical plates $N = 16\,(tR/W)^2$, where W is the baseline peak width; plate height $H = L/N$.
- Resolution $Rs = 2\,(tR2 - tR1)/(W1 + W2)$; Rs of 1.5 is baseline separation.
- Van Deemter equation $H = A + B/u + C{\cdot}u$: A is eddy diffusion (multiple flow paths through packed particles), B/u is longitudinal diffusion (dominant at low linear velocity u), C·u is resistance to mass transfer (dominant at high velocity). Open-tubular capillary GC columns have no A term (Golay equation), which is why they are far more efficient than packed columns; in HPLC, smaller particles (sub-2 µm in UHPLC) shrink A and C, raising efficiency at the price of back-pressure above 600 bar.

The practical rule for choosing a technique: volatile and thermally stable analytes (fatty acid esters, pesticides, flavour volatiles, solvents) go to GC; non-volatile, polar or thermolabile analytes (vitamins, colours, mycotoxins, sugars, antibiotics, carbamate pesticides, acrylamide) go to HPLC or LC-MS/MS; elements go to AAS or ICP-MS.

4.1 High-performance liquid chromatography (HPLC)

Instrumentation

1. Solvent reservoirs and degasser: HPLC-grade solvents (water, acetonitrile, methanol, buffers), filtered through 0.45 or 0.22 µm membranes and degassed (helium sparging or vacuum degasser) to prevent bubbles in the pump and detector.
2. Pump: dual-piston reciprocating pump delivering 0.1 to 10 mL/min against 100 to 400 bar (up to 1200 bar in UHPLC) with pulse dampening; isocratic elution (constant composition) or gradient elution (composition changed with time to elute strongly retained compounds faster) using low-pressure (proportioning valve before one pump) or high-pressure (two pumps) mixing.
3. Injector: six-port rotary valve (Rheodyne) with a fixed loop of 5 to 100 µL, loaded at atmospheric pressure and switched into the high-pressure stream; autosamplers automate this.
4. Column: stainless steel tube, typically 150 to 250 mm long and 4.6 mm internal diameter, packed with 3 to 5 µm porous silica particles chemically bonded with the stationary phase. Reversed-phase columns carry octadecylsilane (ODS, C18) non-polar chains and are used with polar aqueous mobile phases; less polar analytes are retained longer. Normal-phase (bare silica, amino, cyano) retains polar compounds; ion-exchange, size-exclusion, HILIC and chiral phases serve special needs. A guard column protects the analytical column; a column oven (25 to 60 °C) stabilises retention.
5. Detector (see table), data system for integration and quantitation by external standard, internal standard or standard addition.

Detector comparison

Detector	Principle	Sensitivity and selectivity	Gradient compatible	Typical food applications
UV-Vis (fixed or variable wavelength)	Beer-Lambert absorbance at 190 to 700 nm	ng range; selective for chromophores	Yes	Preservatives (benzoate, sorbate at 230 to 254 nm), caffeine (273 nm), vitamins A (325 nm), E (292 nm), D (265 nm), sweeteners, synthetic colours
Photodiode array (PDA, DAD)	Array of diodes records a full spectrum (190 to 800 nm) at every time point	Same as UV plus spectral identification and peak purity	Yes	Synthetic food colours (each has a characteristic visible maximum), multi-vitamin assays, mycotoxins with chromophores, library matching of unknowns
Fluorescence (FLD)	Excitation at one wavelength, emission measured at a longer wavelength at right angles	10 to 1000 times more sensitive than UV; highly selective (few compounds fluoresce)	Yes	Aflatoxins (Ex 365 nm, Em 435 to 450 nm), ochratoxin A, riboflavin, thiamine as thiochrome, PAHs (benzo(a)pyrene), histamine after OPA derivatisation, quinine
Refractive index (RID)	Measures the bulk refractive index difference between column effluent and reference mobile phase	Universal but low sensitivity (µg range); very temperature sensitive	No (baseline drifts with composition)	Sugars (glucose, fructose, sucrose, lactose, maltose), sugar alcohols, ethanol, lipids
Evaporative light scattering (ELSD)	Nebulise, evaporate solvent, scatter light from residual particles	Universal for non-volatile analytes; moderate sensitivity	Yes	Sugars, lipids, triglycerides, saponins
Electrochemical (amperometric, PAD)	Oxidation or reduction at an electrode	Very sensitive for electroactive species	Limited	Ascorbic acid, catechins, sugars by HPAEC-PAD
Conductivity	Ion conductance	Ions	Yes (with suppression)	Ion chromatography of nitrate, nitrite, chloride, sulfite, bromate
Mass spectrometer	Mass-to-charge ratio	Most selective and sensitive	Yes (volatile buffers only)	See LC-MS/MS section

HPSC EXAM TRAP: RID and gradients
The refractive index detector cannot be used with gradient elution because any change in mobile phase composition changes the refractive index and the baseline; it also needs tight temperature control and has the poorest sensitivity. A statement "RID is the detector of choice for trace analysis with gradient HPLC" is wrong on two counts.

Applications

Vitamins: fat-soluble vitamins (A as retinol, D2 and D3, E as tocopherols, K1) are released by alkaline saponification (ethanolic KOH with pyrogallol or BHT antioxidant), extracted into hexane or diethyl ether and determined by reversed-phase HPLC with UV or fluorescence detection; water-soluble B vitamins (B1, B2, B3, B6, folate, B12) are extracted by acid and enzymatic (takadiastase, phosphatase) hydrolysis and separated by ion-pair reversed-phase HPLC with UV, fluorescence (B1 after oxidation to thiochrome, B2 native) or MS detection; vitamin C is extracted in metaphosphoric acid and measured at about 245 nm, with dithiothreitol reduction to include dehydroascorbic acid. Relevant for verifying nutrition labels under the Labelling and Display Regulations and fortification claims under the Fortification Regulations.

Synthetic food colours: India permits eight synthetic colours (Ponceau 4R, Carmoisine, Erythrosine, Tartrazine, Sunset Yellow FCF, Indigo Carmine, Brilliant Blue FCF and Fast Green FCF) with a general ceiling of 100 mg/kg (200 mg/kg in specified products such as certain canned fruits and sweets) singly or in combination, only in listed foods. Non-permitted colours (Metanil Yellow, Rhodamine B, Orange II, Sudan dyes, Auramine, Malachite Green) are frequent adulterants of sweets, spices and beverages. Colours are extracted by adsorption on wool or polyamide, or by solid-phase extraction, and separated by reversed-phase ion-pair HPLC with PDA detection; each colour is identified by retention time plus its visible spectrum, allowing a single run to screen permitted and non-permitted dyes. Screening methods: thin-layer chromatography and spot tests under the FSSAI manual.

Aflatoxins: aflatoxins B1, B2, G1 and G2 (Aspergillus flavus and A. parasiticus in groundnuts, maize, chillies, nuts, copra) and M1 (hydroxylated metabolite of B1 secreted in milk) are extracted with methanol-water or acetonitrile-water, cleaned up on immunoaffinity columns bearing anti-aflatoxin antibodies, eluted with methanol and separated on a C18 column with water-methanol-acetonitrile. B2 and G2 fluoresce strongly in aqueous mobile phase, but B1 and G1 (with a terminal double bond in the dihydrofuran ring) show quenched fluorescence; so post-column derivatisation is applied: reaction with iodine in a heated reactor, electrochemically generated bromine (Kobra cell), or a photochemical reactor (ultraviolet irradiation forming hydrated derivatives), each of which saturates the double bond and restores fluorescence; pre-column trifluoroacetic acid derivatisation is the alternative. Detection: Ex 365 nm, Em 435 to 450 nm. FSSAI limits (Contaminants, Toxins and Residues Regulations as amended): total aflatoxins 15 µg/kg and aflatoxin B1 10 µg/kg in cereals, nuts and oilseeds; aflatoxin M1 0.5 µg/kg in milk. Confirmation and multi-mycotoxin work (ochratoxin A, deoxynivalenol, zearalenone, fumonisins, patulin in apple juice) move to LC-MS/MS.

Sugar profiling: sugars have no chromophore, so they are separated on amino-bonded (NH2) columns with acetonitrile-water (about 75:25) and detected by RID, or on cation-exchange (calcium or lead form) columns with water at 80 °C, or on anion-exchange columns at high pH with pulsed amperometric detection (HPAEC-PAD) for trace sugars and oligosaccharides. Applications: honey authenticity (fructose, glucose, sucrose ratios; detection of added high-fructose syrup), lactose in lactose-free milk, sucrose in fruit juice adulteration, sugar declarations for the nutrition panel and the Indian sugar-content thresholds for high-fat, sugar and salt labelling.

Other routine HPLC targets: preservatives (benzoic and sorbic acid; FSSAI class II preservative limits), sweeteners (saccharin, aspartame, acesulfame-K, sucralose by HPLC-UV or ELSD), caffeine in beverages (FSSAI 145 ppm cap in carbonated beverages, 320 ppm in energy drinks), melamine (HPLC-UV screening, LC-MS/MS confirmation), histamine in fish (fluorescence after derivatisation; relevant to sc

Part 4B. GC, GC-MS, LC-MS/MS & Atomic Spectroscopy

PART 4B: GAS CHROMATOGRAPHY, TANDEM MASS SPECTROMETRY AND ATOMIC SPECTROSCOPY

4.2 Gas chromatography (GC and GC-MS)

4.2.1 First principles

Gas chromatography separates volatile and thermally stable analytes (boiling point generally below about 400 °C, molecular mass below about 600 Da) by partitioning them between an inert gaseous mobile phase, which merely carries the analyte and does not interact with it, and a liquid stationary phase coated on the inner wall of a capillary column. Retention is governed by the vapour pressure of the analyte at the column temperature (hence boiling-point order on non-polar columns) and by the polarity match between analyte and stationary phase. Because retention depends exponentially on temperature, temperature programming (for example 60 °C held 2 minutes, then 10 °C per minute to 300 °C) is used to elute both light and heavy components with good peak shape in one run. Non-volatile analytes (fatty acids, sugars, amino acids, sterols) are first converted to volatile derivatives (methylation, silylation, acylation).

4.2.2 Carrier gas selection

Carrier gas	Optimum linear velocity	Efficiency and speed	Cost and safety	Comment
Helium	20 to 40 cm/s	Good efficiency, fast analyses, flat van Deemter curve	Expensive, supply shortages, inert	Default for GC-MS (ionisation potential of 24.6 eV, no interference) and for most capillary work
Hydrogen	40 to 60 cm/s	Fastest analyses, flattest curve, lowest viscosity	Cheap (generated on site by electrolysis), flammable (explosive 4 to 75 percent in air), can hydrogenate some analytes in the injector	Increasingly used with FID and modern MS systems with leak sensors
Nitrogen	10 to 15 cm/s	Highest efficiency but only at a slow velocity, long run times	Cheapest, safe	Common with packed columns and FID in Indian laboratories; unsuitable for MS because of lower pumping efficiency and ion source effects

Carrier gas purity must be at least 99.999 percent with oxygen and moisture traps, because oxygen oxidises the stationary phase at high temperature (column bleed), and electron capture detectors require oxygen-free gas. Make-up gas (nitrogen or argon-methane for ECD) is added at the detector to optimise response.

4.2.3 Columns

- Packed columns (2 to 4 m, 2 to 4 mm internal diameter, support such as Chromosorb coated with 3 to 10 percent liquid phase) are now historical except for permanent gases and some pharmacopoeial methods.
- Capillary columns (wall-coated open tubular, WCOT) of fused silica with a polyimide outer coating: typically 30 m long (15 to 100 m), 0.25 mm internal diameter (0.1 to 0.53 mm), 0.25 µm film thickness. They give 100000 to 300000 theoretical plates and are the basis of all modern food methods.

Stationary phase (trade names)	Polarity	Elution order	Food applications
100 percent dimethylpolysiloxane (DB-1, HP-1, OV-1)	Non-polar	Boiling point	Hydrocarbons, mineral oil, general screening
5 percent phenyl, 95 percent dimethylpolysiloxane (DB-5, HP-5MS, Rtx-5)	Low polarity	Boiling point with slight aromatic selectivity	Multi-residue pesticides (organochlorines, organophosphates, pyrethroids), polycyclic aromatic hydrocarbons, sterols, phthalates; the universal GC-MS column
50 percent phenyl, 50 percent methylpolysiloxane (DB-17, HP-50)	Intermediate	Polarity and boiling point	Triglycerides, steroids, drug residues
Polyethylene glycol, Carbowax 20M (DB-WAX, HP-INNOWax, Supelcowax)	Polar	Hydrogen-bonding and polarity	Flavour volatiles (alcohols, esters, aldehydes), fatty acid methyl esters (general profile), ethanol and methanol in alcoholic beverages, solvents
Highly polar biscyanopropyl polysiloxane, 100 m (SP-2560, CP-Sil 88, HP-88)	Very polar	Separates positional and geometric isomers; trans isomers elute before the corresponding cis isomers	Trans fatty acid determination (AOAC 996.06, AOCS Ce 1h-05), conjugated linoleic acid

The Kovats retention index (normalising retention to that of n-alkanes) allows transfer of flavour identifications between laboratories. Column bleed, indicated by rising baseline at high temperature, is minimised by using low-bleed MS-grade phases.

4.2.4 Injection: split versus splitless

The sample (0.5 to 2 µL of liquid) is vaporised in a heated injector (200 to 300 °C) lined with a deactivated glass liner.

Mode	How it works	Use	Trade-off
Split	Only a fraction of the vapour (split ratio 1:20 to 1:200) enters the column; the rest is vented	Concentrated samples (flavour concentrates, FAMEs, edible oil adulteration), neat liquids	Sharp bands, no column overloading, but 95 to 99 percent of the sample is discarded, so sensitivity is low

Mode	How it works	Use	Trade-off
Splitless	Split vent is closed for 30 to 90 seconds so nearly the entire vapour is transferred to a cool column (held 10 to 20 °C below the solvent boiling point to refocus the analytes by the solvent effect); the vent then opens to purge residual solvent	Trace analysis: pesticide residues, dioxins, contaminants at µg/kg	Maximum sensitivity but broader solvent peak, risk of discrimination of high boilers and matrix build-up in the liner
On-column	Liquid deposited directly in the column at low temperature	Thermally labile analytes, triglycerides	No discrimination, but column contamination
Programmed-temperature vaporiser (PTV) and large-volume injection	Cold injection then rapid heating	Ultra-trace pesticides	Complex optimisation
Static or dynamic headspace, purge-and-trap, solid-phase microextraction (SPME)	Only the vapour above the sample is injected, or volatiles are adsorbed on a coated fibre	Flavour and off-flavour volatiles, residual solvents, ethylene oxide, acetaldehyde in PET water	Solvent-free, matrix-free, but equilibrium dependent

4.2.5 Detectors

Detector	Principle	Selectivity	Detection limit and linear range	Destructive	Key food applications
Flame ionisation detector (FID)	Column effluent burns in a hydrogen-air flame; carbon-containing molecules form CHO+ ions and electrons, collected at a polarised electrode as a current	Near universal for organic carbon; response roughly proportional to the number of oxidisable carbon atoms (equal-carbon response); no response to water, carbon dioxide, carbon monoxide, formic acid, ammonia, inorganic gases, so aqueous and air samples give clean baselines	About 10^-12 g carbon per second; linear over 10^7, the widest of any detector	Yes	FAMEs and trans fat profile, ethanol and methanol in liquor, sugars and sterols after silylation, flavour compounds, residual solvents

Detector	Principle	Selectivity	Detection limit and linear range	Destructive	Key food applications
Electron capture detector (ECD)	A 63Ni foil (beta emitter, about 15 mCi, half-life 100 years) ionises the make-up gas (nitrogen or argon with 5 percent methane) producing a standing current of thermal electrons; electronegative analytes (halogens, nitro groups, conjugated carbonyls) capture electrons, reducing the current	Highly selective for halogenated compounds; insensitive to hydrocarbons, alcohols, amines	Femtogram level for polyhalogenated compounds (most sensitive GC detector for organochlorines); narrow linear range, only 10^3 to 10^4, so dilution and bracketing calibration are needed	No (non-destructive)	Organochlorine pesticides (DDT, HCH isomers, endosulfan, aldrin, dieldrin, heptachlor, chlordane), polychlorinated biphenyls, chlorinated solvents, trichloroanisole cork taint; the source is radioactive and requires Atomic Energy Regulatory Board licensing and periodic wipe tests in India
Nitrogen-phosphorus detector (NPD, thermionic)	Heated rubidium or caesium silicate bead in a low-hydrogen plasma	Nitrogen and phosphorus compounds, 10^4 times more selective than FID	pg range	Yes	Organophosphorus (chlorpyrifos, malathion, monocrotophos, dimethoate) and carbamate pesticides, nitrosamines
Flame photometric detector (FPD)	Hydrogen-rich flame excites sulfur (S2 emission at 394 nm) and phosphorus (HPO emission at 526 nm)	Sulfur or phosphorus	pg range; sulfur response is quadratic	Yes	Organophosphorus pesticides, sulfur volatiles in onion, garlic, beer
Thermal conductivity detector (TCD)	Change in thermal conductivity of the carrier gas across a heated filament (Wheatstone bridge)	Universal including inorganic gases	ng range (low)	No	Headspace gases in modified atmosphere packs (oxygen, carbon dioxide, nitrogen), water, permanent gases
Mass spectrometer (MS)	Electron ionisation at 70 eV, mass analysis of fragments	Universal and structurally specific	pg range in selected ion monitoring	Yes	Confirmation of pesticides, flavour identification, adulterants, packaging migrants (benzophenone, phthalates, mineral oil)

HPSC EXAM TRAP: FID versus ECD
FID is the universal carbon detector with the widest linear range and is destructive. ECD is the selective halogen detector, uses a 63Ni beta source, is the most sensitive for organochlorine pesticides, is non-

destructive, and has the narrowest linear range. A statement that ECD responds to all organic compounds, or that FID is used for organochlorine residue screening at µg/kg, is wrong.

4.2.6 GC-MS fundamentals

- Interface: the capillary column runs directly into the ion source held under vacuum (10^-5 to 10^-6 torr) maintained by a turbomolecular pump; helium is pumped away.
- Electron ionisation (EI): a heated tungsten or rhenium filament emits electrons accelerated through 70 V; collision ejects an electron from the analyte, M + e- → M+• + 2e-, producing the molecular ion, which fragments into characteristic ions. 70 eV is used because ionisation efficiency reaches a plateau there, so small fluctuations do not alter the spectrum and spectra are reproducible between instruments and comparable with libraries (NIST, Wiley, with more than 300000 spectra). Fragmentation follows predictable rules (alpha-cleavage, McLafferty rearrangement, tropylium ion at m/z 91 for benzyl compounds).
- Chemical ionisation (CI) with methane, isobutane or ammonia reagent gas is a soft technique giving the protonated molecule [M+H]+ when the molecular ion is absent under EI.
- Analysers: single quadrupole (most common), ion trap, time-of-flight (high resolution, exact mass), triple quadrupole GC-MS/MS for pesticide residues with MRM as described for LC-MS/MS.
- Acquisition modes: full scan (for identification, library search of unknowns) and selected ion monitoring (SIM, where the quadrupole dwells on two to four characteristic ions per analyte, raising sensitivity 10 to 100 fold; one quantifier ion and qualifier ions whose ratios must match the standard within the tolerances of the EU SANTE pesticide guidance, usually within 30 percent relative).

4.2.7 Derivatisation: fatty acid methyl esters (FAMEs)

Triglycerides are non-volatile and free fatty acids tail badly on GC columns because of hydrogen bonding, so lipids are converted to volatile fatty acid methyl esters by transesterification. Two official routes:

1. Boron trifluoride-methanol (AOAC 969.33, IS 548 Part 3, AOCS Ce 2-66): the fat is first saponified by refluxing with 0.5 N methanolic sodium hydroxide (triglyceride + 3NaOH → glycerol + 3RCOONa), then 12 to 14 percent BF3 in methanol is added and the mixture refluxed at 100 °C for 2 to 5 minutes; the Lewis acid catalyses esterification, RCOOH + CH3OH → RCOOCH3 + H2O, giving quantitative methylation of all acids including free acids and phospholipid acids. Esters are extracted into heptane or hexane after adding saturated sodium chloride solution.
2. Methanolic potassium hydroxide (ISO 12966-2, IUPAC 2.301, rapid cold method): 2 N KOH in methanol is shaken with the oil dissolved in heptane at room temperature for 1 to 2 minutes; base-catalysed transesterification, RCOOR' + CH3OH → RCOOCH3 + R'OH, proceeds via the methoxide ion. It is fast and avoids heating (no isomerisation of conjugated or polyunsaturated acids) but does not methylate free fatty acids and fails for oils with high acidity.

Alternatives: 2 percent sulfuric acid in methanol (acid catalysed, slow), sodium methoxide, trimethylsulfonium hydroxide (one-step in the injector).

Analysis: 100 m biscyanopropyl polysiloxane column, FID, hydrogen or helium carrier, oven 140 to 240 °C programme, split injection, internal standard C17:0 or C23:0 methyl ester, identification by comparison with a 37-component FAME mix; results expressed as percent of total fatty acids. On the polar column, elution within each carbon number is saturated, then trans, then cis isomers, so elaidic acid (C18:1 trans-9) elutes before oleic acid (C18:1 cis-9), and the sum of the trans peaks gives total trans fatty acids.

Regulatory linkage: FSSAI limits industrially produced trans fatty acids to 2 percent by weight of total fats in all fats and oils and in foods containing fats and oils (effective 1 January 2022, down from 5 percent in 2021 and 10 percent earlier); a trans fat free claim requires not more than 0.2 g per 100 g or 100 mL. The same FAME profile detects adulteration (for example argemone-free but cheaper palm olein in groundnut oil, lard in ghee by Reichert-Meissl and butyric acid C4:0 content, which only milk fat contains at about 3 to 4 percent) and verifies omega-3 claims.

4.2.8 Pesticide residue screening by GC

Sample preparation uses the QuEChERS method (quick, easy, cheap, effective, rugged, safe; AOAC 2007.01, EN 15662): 10 g homogenised produce is shaken with 10 mL acetonitrile, salted out with anhydrous magnesium sulfate and sodium chloride (or citrate buffers), and the extract is cleaned by dispersive solid-phase extraction with primary secondary amine (PSA) sorbent, which removes fatty acids and sugars, plus C18 for fat and graphitised carbon black for chlorophyll. Organochlorines are screened by GC-ECD, organophosphates by GC-NPD or FPD, and all are confirmed by GC-MS or GC-MS/MS. The FSSAI default maximum residue limit for any pesticide not specifically listed in the Contaminants, Toxins and Residues Regulations is 0.01 mg/kg, and DDT, endosulfan, lindane and many organochlorines are banned or restricted under the Insecticides Act, 1968, so their presence at any level is an enforcement issue.

4.3 Tandem mass spectrometry (LC-MS/MS)

4.3.1 Why LC-MS/MS exists

Most veterinary drugs, mycotoxins, modern pesticides, acrylamide and melamine are polar, thermally labile or non-volatile, so they are unsuited to GC without derivatisation, and in complex food matrices a UV detector cannot distinguish a residue at 1 µg/kg from co-eluting matrix. Coupling HPLC to a triple quadrupole mass spectrometer solves both problems: the liquid chromatograph handles polar molecules, and two stages of mass filtering give selectivity so high that clean-up can be minimal while sensitivity reaches sub-µg/kg.

4.3.2 Atmospheric pressure ionisation: ESI versus APCI

The challenge is to convert ions in a liquid flowing at 0.2 to 1 mL per minute into gas-phase ions at atmospheric pressure, then sample them into the vacuum.

Electrospray ionisation (ESI, John Fenn, Nobel Prize 2002): the eluent passes through a stainless steel capillary held at 3 to 5 kV relative to the counter electrode; the field charges the liquid surface, draws it into a Taylor cone and sprays a fine mist of highly charged droplets, assisted by a coaxial nitrogen nebulising gas. Heated drying gas evaporates solvent, droplets shrink until charge repulsion exceeds surface tension (the Rayleigh limit) and they undergo repeated Coulomb fission; finally ions leave the droplets (ion evaporation model for small molecules; charged residue model for proteins). Ionisation happens in solution, so ESI favours compounds that are already ionic or readily protonated or deprotonated: acids, bases, zwitterions, polar drugs, peptides, proteins (which acquire multiple charges). Positive mode gives [M+H]+ (or [M+Na]+, [M+NH4]+), negative mode gives [M-H]-. ESI is the softest ionisation (no fragmentation, intact molecular information) but is prone to matrix effects: co-eluting matrix components compete for charge and suppress (or enhance) the analyte signal, which is why isotope-labelled internal standards (13C or deuterium) and matrix-matched calibration are mandatory in residue methods.

Atmospheric pressure chemical ionisation (APCI): the eluent is nebulised and vaporised in a heated tube at 350 to 500 °C; a corona discharge needle (about 3 kV) ionises solvent and nitrogen to form reagent ions (H_3O^+, $CH_3OH_2^+$), which transfer a proton to the analyte in the gas phase ($M + H_3O^+ \rightarrow [M+H]^+ + H_2O$). Because ionisation occurs in the gas phase, APCI suits neutral, moderately polar to non-polar, thermally stable molecules of low to medium mass (below about 1500 Da) and gives only singly charged ions; it tolerates higher flow rates and non-polar mobile phases and is less affected by matrix suppression, but cannot be used for thermolabile or very polar or large molecules. Atmospheric pressure photoionisation (APPI, krypton lamp 10 eV) extends the range to non-polar compounds such as polycyclic aromatic hydrocarbons and vitamins D and K.

Criterion	ESI	APCI
Where ionisation occurs	In solution, within charged droplets	In the gas phase, by proton transfer from reagent ions
Energy source	High voltage on the spray capillary	Corona discharge needle after thermal vaporisation
Best analytes	Polar, ionic, basic or acidic, thermally labile, high mass (proteins, peptides, tetracyclines, aminoglycosides, glyphosate)	Neutral, less polar, thermally stable, low mass (steroids, triglycerides, carbamates, some mycotoxins, vitamins)
Charge state	Multiple charges possible	Singly charged
Flow rate tolerance	nL to 1 mL per minute	0.2 to 2 mL per minute
Matrix effect	High (ion suppression)	Lower
Fragmentation	Minimal (softest)	Minimal, slightly more thermal breakdown
Share of food residue methods	About 80 to 90 percent	10 to 20 percent

4.3.3 Triple quadrupole (QqQ) architecture

A quadrupole is four parallel metal rods to which a direct current voltage and a radio-frequency voltage are applied; for a given voltage ratio only ions of one mass-to-charge ratio (m/z) have stable trajectories and pass through, while others strike the rods (described by the Mathieu equations). Scanning the voltages sweeps the transmitted mass. The triple quadrupole arranges three such devices in line:

1. Q1 (first mass filter): set to transmit only the precursor ion of the target analyte, for example [M+H]+ at m/z 445 for tetracycline. Everything else from the ion source, including matrix ions of other masses, is rejected here.
2. q2 (collision cell): a radio-frequency-only quadrupole or hexapole (no mass filtering; lower-case q) filled with argon or nitrogen at about 1 to 5 millitorr. The precursor ion is accelerated into the gas (collision energy 5 to 50 eV) and undergoes collision-induced dissociation (CID): kinetic energy is converted to internal energy and the ion breaks at its weakest bonds into product ions and neutral fragments.
3. Q3 (second mass filter): set to transmit only a specific product ion, for example m/z 410 (loss of ammonia and water from tetracycline). Only an ion that had the correct precursor mass and also produced the correct fragment reaches the detector (electron multiplier), so chemical noise is almost eliminated.

Scan modes derived from this architecture: product ion scan (Q1 fixed, Q3 scanned; used for method development and structure elucidation), precursor ion scan (Q3 fixed, Q1 scanned; finds all compounds giving a common fragment, for example all glucuronides), neutral loss scan (both scanned with a fixed offset), and multiple reaction monitoring.

HPSC EXAM TRAP: q2 is not a mass analyser
In a triple quadrupole the precursor is selected in Q1, fragmented in the collision cell q2 (which only transmits ions and does no mass selection), and the product is selected in Q3. A statement that the collision cell selects the precursor or product ion is wrong.

4.3.4 Multiple reaction monitoring (MRM, also called selected reaction monitoring)

In MRM both Q1 and Q3 are fixed on a precursor-product pair called a transition, with a dwell time of 5 to 100 milliseconds, then switched to the next transition; a modern instrument monitors several hundred transitions in one run using retention-time windows (scheduled MRM). For each analyte at least two transitions are acquired: the most intense (quantifier) for calibration and a second (qualifier) whose ratio

to the first must match the reference standard within tolerance (EU Decision 2002/657/EC and SANTE guidance). Identification points: a precursor ion earns 1.0 point and each product ion 1.5 points, so one precursor with two products gives 4 points, which is the minimum required for confirming banned (Group A) substances such as chloramphenicol and nitrofurans; permitted drugs (Group B) need 3 points. Advantages of MRM: selectivity (two-stage filtering), sensitivity (the detector sees only the target ion, giving sub-µg/kg limits), speed (no derivatisation), and the ability to quantify hundreds of residues simultaneously.

4.3.5 Applications with transitions you can quote

Target	Why it matters	Ionisation and typical transitions (m/z)	Statutory reference
Multi-residue pesticides (300 to 600 compounds in one injection after QuEChERS)	Carbamates, neonicotinoids, triazoles, strobilurins, phenylureas and modern polar pesticides are not amenable to GC	ESI positive, for example carbendazim 192 → 160, imidacloprid 256 → 209, acephate 184 → 143	FSSAI Contaminants, Toxins and Residues Regulations, default 0.01 mg/kg; national monitoring programme on pesticide residues
Chloramphenicol	Banned broad-spectrum antibiotic (aplastic anaemia), zero tolerance; found in shrimp, honey, milk	ESI negative, 321 → 152 (quantifier) and 321 → 257; isotope internal standard chloramphenicol-d5	Minimum required performance limit 0.3 µg/kg (EU), prohibited in India for food animals; shrimp export rejections
Nitrofuran metabolites (AOZ, AMOZ, AHD, SEM)	Banned; metabolites are tissue-bound, so acid hydrolysis and derivatisation with 2-nitrobenzaldehyde precede LC-MS/MS	ESI positive, AOZ derivative 236 → 134	MRPL 1 µg/kg; India banned nitrofurans in aquaculture (2002)
Tetracyclines (tetracycline, oxytetracycline, chlortetracycline, doxycycline)	Widely used in poultry, aquaculture and honey; residues cause allergy, resistance, tooth discolouration	ESI positive, tetracycline 445 → 410 and 445 → 427; oxytetracycline 461 → 426 and 461 → 443; extraction with EDTA-McIlvaine buffer because tetracyclines chelate metals	FSSAI tolerance limits 0.1 mg/kg in muscle, 0.3 in liver, 0.6 in kidney of food animals, 0.1 mg/kg in milk; honey must be free of antibiotics for export
Sulfonamides, fluoroquinolones, beta-lactams, macrolides	Permitted drugs with withdrawal periods; honey and shrimp exports	ESI positive, for example sulfamethazine 279 → 186, enrofloxacin 360 → 316	FSSAI Schedule of permitted veterinary drugs and tolerance limits; EU Regulation 37/2010 for export
Acrylamide	Probable human carcinogen (IARC 2A) formed above 120 °C by the Maillard reaction between free asparagine and reducing sugars in fried potato, biscuits, bread crust and roasted coffee	ESI positive, 72 → 55 (loss of ammonia) quantifier and 72 → 44 qualifier, internal standard 13C3-acrylamide (75 → 58); water extraction, Carrez clarification, solid-phase extraction; alternative GC-MS after bromination to 2,3-dibromopropionamide	EU benchmark levels (Regulation 2017/2158): potato crisps 750 µg/kg, French fries 500, soft bread 50, biscuits 350, roasted coffee 400; FSSAI has issued guidance on mitigation (lower frying temperature, asparaginase, blanching)
Melamine	Nitrogen-rich adulterant (66.6 percent nitrogen) added to inflate Kjeldahl protein; 2008 infant formula crisis; forms kidney stones with cyanuric acid	ESI positive, 127 → 85 and 127 → 68; hydrophilic interaction chromatography (HILIC)	Limits adopted in India following Codex: 1 mg/kg in powdered infant formula, 0.15 mg/kg in liquid infant formula, 2.5 mg/kg in other foods and feeds

Target	Why it matters	Ionisation and typical transitions (m/z)	Statutory reference
Mycotoxins (aflatoxins, ochratoxin A, deoxynivalenol, zearalenone, fumonisins, patulin)	Multi-mycotoxin methods replace single HPLC-FLD methods	ESI positive and negative switching, aflatoxin B1 313 → 285 and 313 → 241	FSSAI mycotoxin limits as in Section 4.1

4.4 Atomic spectroscopy for heavy metals

4.4.1 First principles

Atomic absorption relies on free, gaseous, ground-state atoms absorbing light at a resonance wavelength characteristic of the element (lead 283.3 nm, cadmium 228.8 nm, arsenic 193.7 nm, mercury 253.7 nm, copper 324.8 nm, zinc 213.9 nm). According to the Boltzmann distribution, at flame temperatures (2000 to 3000 K) more than 99.9 percent of atoms of most elements are in the ground state, so absorption is far more sensitive and matrix independent than flame emission for all but the easily excited alkali metals (sodium, potassium, lithium, measured by flame photometry). The instrument comprises a hollow cathode lamp of the same element as the analyte (emitting a very narrow line, which is why one lamp per element is needed and AAS is sequential), an atomiser (flame, furnace, quartz cell), a monochromator to isolate the resonance line, a photomultiplier detector, and a background corrector (deuterium continuum lamp or Zeeman effect using a strong magnetic field to split the absorption line). Absorbance follows the Beer-Lambert law, $A = \varepsilon \cdot b \cdot c$, and quantitation uses external calibration, standard addition for difficult matrices, and certified reference materials for validation.

Sample preparation is common to all techniques and is the usual source of error. Wet digestion with concentrated nitric acid and hydrogen peroxide in closed-vessel microwave digesters (180 to 200 °C) is the method of choice because it is fast and retains volatile elements; dry ashing at 450 to 550 °C followed by dissolution in dilute nitric acid is cheaper but loses mercury entirely and partly loses arsenic, cadmium, lead and selenium by volatilisation and retention in silica. All glassware is acid-washed; reagent blanks are run with every batch.

4.4.2 Flame AAS (FAAS)

The sample solution is drawn by a pneumatic nebuliser into a spray chamber, where only the finest 5 to 10 percent of the aerosol reaches the burner; the rest drains to waste. An air-acetylene flame (about 2300 °C) through a 10 cm slot burner atomises most elements; the hotter nitrous oxide-acetylene flame (about 2900 °C) is needed for refractory, oxide-forming elements (aluminium, calcium at high concentration, silicon, titanium, vanadium). Atom residence time in the light path is only a few milliseconds. Consequences: detection limits in the mg/L (ppm) to high µg/L range (lead about 0.02 mg/L, cadmium about 0.002 mg/L), precision of 0.5 to 1 percent, analysis in 10 seconds per element, low cost, robust. Interferences: chemical (phosphate or sulfate forming refractory calcium compounds, overcome by adding lanthanum chloride as a releasing agent), ionisation (alkali and alkaline earth metals ionise in hot flames, suppressed by adding excess caesium or potassium), spectral (rare, because lines are narrow), and background absorption or scatter from salts (corrected by the deuterium lamp). FAAS is adequate for nutrients and additives (iron, zinc, copper, calcium, sodium, potassium in fortified foods, tin in canned foods at 250 mg/kg) but usually too insensitive for lead, cadmium and arsenic at the µg/kg limits now set for foods.

4.4.3 Graphite furnace AAS (GFAAS, electrothermal AAS, ETAAS)

A 10 to 50 µL aliquot is pipetted onto a L'vov platform inside a pyrolytically coated graphite tube, which is heated electrically under an argon purge (argon prevents the tube from burning) through a programme:

drying at 100 to 120 °C (30 seconds) to evaporate solvent; pyrolysis or ashing at 400 to 1200 °C to destroy organic matter and volatilise matrix salts without losing the analyte; atomisation at 1800 to 2800 °C within 1 to 2 seconds, with gas flow stopped so the atom cloud is confined in the light path; and a clean-out step at 2600 °C. Matrix modifiers are added to allow a higher pyrolysis temperature: palladium with magnesium nitrate stabilises arsenic, selenium and lead; ammonium dihydrogen phosphate converts volatile cadmium and lead chlorides to less volatile phosphates. Because the entire sample is atomised into a confined volume and atoms reside for seconds rather than milliseconds, sensitivity improves 100 to 1000 fold: detection limits of 0.01 to 1 µg/L (ppb), for example lead 0.05 µg/L and cadmium 0.002 µg/L. Drawbacks: slow (2 to 3 minutes per determination, single element), strong background from matrix smoke requiring Zeeman correction, more chemical interference, higher cost, limited linear range, and need for the stabilised temperature platform furnace (STPF) protocol for reliable results. GFAAS is the workhorse for lead, cadmium and chromium in FSSAI contaminant testing where ICP-MS is unavailable.

Criterion	Flame AAS	Graphite furnace AAS
Atomiser	Air-acetylene or nitrous oxide-acetylene flame, 2300 to 2900 °C	Electrically heated graphite tube, up to 2800 °C under argon
Sample use	Continuous aspiration, 1 to 5 mL per minute, 5 to 10 percent efficiency	10 to 50 µL, essentially 100 percent atomised
Residence time	Milliseconds	Seconds
Detection limit	mg/L to 0.01 mg/L (ppm)	µg/L to 0.01 µg/L (ppb), 100 to 1000 times better
Analysis time	10 seconds	2 to 3 minutes
Precision	0.5 to 1 percent RSD	1 to 5 percent RSD
Interference and background	Low	High; Zeeman correction and matrix modifiers essential
Cost	Low	Moderate
Typical food use	Nutrient minerals, tin in cans, zinc, copper, iron	Lead, cadmium, chromium, nickel, arsenic (after modifier) at µg/kg

4.4.4 Hydride generation AAS (HGAAS)

Arsenic, selenium, antimony, bismuth, tellurium, tin and germanium have resonance lines in the far ultraviolet (arsenic 193.7 nm, selenium 196.0 nm) where the flame itself absorbs strongly, so direct FAAS is insensitive. These elements form volatile covalent hydrides when their lower oxidation states are reduced by sodium borohydride in hydrochloric acid:

$NaBH_4 + 3H_2O + HCl \rightarrow H_3BO_3 + NaCl + 8H$ (nascent hydrogen)

$As^{3+} + 3H \rightarrow AsH_3$ (arsine); $Se^{4+} + 4H \rightarrow SeH_2$ (hydrogen selenide); $Sb^{3+} + 3H \rightarrow SbH_3$ (stibine)

The gaseous hydride is swept by argon from the reaction vessel (or a continuous-flow manifold with a gas-liquid separator) into a quartz tube held at 900 to 1000 °C in the light path, where it decomposes to free atoms ($2AsH_3 \rightarrow 2As + 3H_2$). Because the analyte is separated from the digest matrix and introduced as a gas with near 100 percent efficiency, sensitivity improves 10 to 100 fold over FAAS, with detection limits of 0.01 to 0.1 µg/L. Two chemistry points are routinely examined: the analyte must be in the lower oxidation state, so arsenic(V) is pre-reduced to arsenic(III) with potassium iodide and ascorbic acid, and selenium(VI) is pre-reduced to selenium(IV) by heating in 6 M hydrochloric acid; and transition metals (copper, nickel) interfere by consuming borohydride or decomposing the hydride, controlled by masking agents or dilution. HGAAS after digestion gives total inorganic arsenic; it does not respond to arsenobetaine, the dominant and non-toxic arsenic form in fish and shrimp, so a low HGAAS result on seafood is not a

contradiction of a high total arsenic by ICP-MS. Arsenic speciation (arsenite, arsenate, methylated forms) requires HPLC coupled to ICP-MS and is relevant for rice, where FSSAI and Codex limit inorganic arsenic (Codex 0.2 mg/kg in polished rice, 0.35 mg/kg in husked rice) and drinking water is limited to 0.01 mg/L under IS 10500.

4.4.5 Cold vapour AAS (CVAAS) for mercury

Mercury is unique among metals in having an appreciable vapour pressure of the free monatomic element at room temperature, so no flame or furnace is required. After oxidative digestion (nitric and sulfuric acids with potassium permanganate or persulfate, in closed vessels to prevent loss, with excess oxidant destroyed by hydroxylamine hydrochloride just before measurement), mercury(II) is reduced to the metal with acidic stannous chloride:

$Hg^{2+} + Sn^{2+} \rightarrow Hg^0 + Sn^{4+}$ (alternatively with sodium borohydride: $Hg^{2+} + 2BH_4^- + 6H_2O \rightarrow Hg^0 + 2H_3BO_3 + 7H_2$)

A stream of argon or air sweeps the mercury vapour through a drying tube into a long-path quartz absorption cell at room temperature, where absorbance is measured at the 253.7 nm resonance line from a mercury hollow cathode or electrodeless discharge lamp. Detection limits are 0.01 to 0.1 µg/L, improved a further 10 to 100 fold by pre-concentrating the vapour on a gold trap (amalgamation) and releasing it by heating. Cold vapour atomic fluorescence (CVAFS) is even more sensitive. Precautions: mercury adsorbs on surfaces and is lost on dry ashing or open-vessel heating; samples with high chloride can lose mercury as volatile chlorides; the technique measures total mercury, while the toxicologically relevant species in fish is methylmercury (about 90 percent of fish mercury, determined by GC-ECD, GC-AFS or LC-ICP-MS). FSSAI limits mercury in fish and fishery products at 0.5 mg/kg (1.0 mg/kg in listed predatory fish such as tuna, shark and swordfish) and methylmercury at 0.25 mg/kg, and mercury in other foods at 1.0 mg/kg under the Contaminants, Toxins and Residues Regulations, 2011, as amended.

4.4.6 Inductively coupled plasma mass spectrometry (ICP-MS)

Components and mechanism:

1. Sample introduction: a peristaltic pump feeds the digested solution (total dissolved solids kept below 0.2 percent) to a concentric or cross-flow nebuliser; a spray chamber passes only the finest aerosol (1 to 2 percent) to the plasma.
2. Torch and plasma: three concentric quartz tubes carry argon as plasma (outer, cooling) gas at 12 to 17 L per minute, auxiliary gas and nebuliser gas. A two- or three-turn copper load coil around the torch is driven by a radio-frequency generator at 27.12 MHz (some instruments use 40.68 MHz) at 1.0 to 1.6 kW. A Tesla spark seeds free electrons; the oscillating magnetic field accelerates them, their collisions ionise argon, and the resulting eddy currents sustain a self-heating, electrodeless plasma at 6000 to 10000 K (the analytical zone is about 6500 to 8000 K). The aerosol is desolvated, vaporised, atomised and ionised within a few milliseconds. Because the first ionisation energy of argon is 15.76 eV, every element with a lower first ionisation energy is ionised efficiently, more than 90 percent for most metals; only helium, neon and fluorine are not ionised, and elements with high ionisation energies (arsenic 9.8 eV, mercury 10.4 eV, selenium 9.75 eV) are ionised 30 to 50 percent, which still allows ng/L detection.
3. Interface: the plasma, at atmospheric pressure, is sampled through a water-cooled nickel or platinum sampler cone (orifice about 1 mm) into a region at a few torr, then through a skimmer cone (orifice about 0.4 mm) into the high vacuum; the supersonic expansion freezes the ion population.
4. Ion optics and collision-reaction cell: electrostatic lenses focus the ions and remove photons and neutrals; a collision-reaction cell filled with helium (kinetic energy discrimination) or a reactive gas (hydrogen, ammonia, oxygen) removes polyatomic interferences formed from argon, water and acids.
5. Mass analyser and detector: a quadrupole scans the full mass range 2 to 260 in less than a second, so all elements are determined simultaneously (quasi-simultaneously by fast sequential scanning); high-resolution sector field and triple quadrupole ICP-MS/MS variants resolve the most difficult interferences. A dual-mode electron multiplier covers nine orders of magnitude, from ng/L to hundreds of mg/L.

Performance: detection limits of 0.001 to 0.1 µg/L (sub-ppt to ppt) for lead, cadmium, mercury, arsenic, uranium and the rare earths; 70 or more elements in two to three minutes; isotope ratio measurement (lead isotopes for source attribution, enriched stable isotopes for bioavailability studies); and coupling to HPLC or GC for speciation (inorganic arsenic in rice, methylmercury in fish, chromium(VI)). Internal standards (rhodium, indium, bismuth, germanium) correct for drift and matrix suppression. Interferences the examiner expects you to name: isobaric overlaps (114Sn on 114Cd, 204Hg on 204Pb, corrected by equations or alternative isotopes 111Cd and 208Pb) and polyatomic ions ($^{40}Ar^{35}Cl^+$ on 75As, the classic problem in high-chloride seafood digests; $^{40}Ar^{12}C^+$ on 52Cr; $^{40}Ar^{16}O^+$ on 56Fe; $^{40}Ar^{40}Ar^+$ on 80Se; $^{35}Cl^{16}O^+$ on 51V), removed by the collision cell or by using hydrogen as reaction gas. Mercury shows a memory effect from adsorption in the sample path, controlled by adding gold (which forms a soluble mercury-gold complex) to all solutions. Limitations: capital and running cost (argon consumption about 15 to 20 L per minute), dissolved solids limit, the need for clean-room practices because the sensitivity makes laboratory contamination the dominant error. For comparison, ICP optical emission spectrometry (ICP-OES) uses the same plasma but measures emitted light, with detection limits in the µg/L range, suited to nutrient minerals rather than ultra-trace contaminants.

Technique	Atomisation or ionisation source	Temperature	Elements	Detection limit (solution)	Throughput	Typical FSSAI use
Flame AAS	Chemical flame	2300 to 2900 °C	One at a time, about 65 metals	mg/L to 0.01 mg/L	10 seconds per element	Tin in canned food (250 mg/kg), iron, zinc, copper, calcium in fortified foods
Graphite furnace AAS	Resistively heated graphite tube	up to 2800 °C	One at a time	µg/L to 0.01 µg/L	2 to 3 minutes per element	Lead, cadmium, chromium, nickel at µg/kg limits
Hydride generation AAS	Chemical reduction with sodium borohydride, heated quartz tube	900 to 1000 °C	Arsenic, selenium, antimony, bismuth, tellurium, tin	0.01 to 0.1 µg/L	1 to 2 minutes	Total inorganic arsenic in rice, water, spices; selenium
Cold vapour AAS	Reduction with stannous chloride, room-temperature quartz cell at 253.7 nm	Ambient	Mercury only	0.01 to 0.1 µg/L (lower with gold amalgamation)	1 to 2 minutes	Total mercury in fish, shellfish, rice
ICP-MS	Argon inductively coupled plasma ionisation, quadrupole mass analysis	6000 to 10000 K	70 or more elements simultaneously, isotopes	0.001 to 0.1 µg/L (ppt)	2 to 3 minutes for all elements	Lead, cadmium, arsenic, mercury multielement screening, speciation, infant food, spices, export certification

HPSC EXAM TRAP: match the element to the technique
Mercury is measured by cold vapour AAS at 253.7 nm after stannous chloride reduction, with no flame.

Arsenic and selenium are measured by hydride generation AAS after sodium borohydride reduction. Lead and cadmium at µg/kg need graphite furnace AAS or ICP-MS, not flame AAS. ICP-MS uses argon plasma at 6000 to 10000 K and a quadrupole mass analyser; it is not an absorption technique and uses no hollow cathode lamp. A statement pairing mercury with hydride generation, or arsenic with cold vapour, is wrong.

Part 5. Proximate Analysis (AOAC/FSSAI Methods)

PART 5: OFFICIAL PROXIMATE ANALYSIS PROTOCOLS (AOAC AND FSSAI MANUAL METHODS)

5.0 The Weende system

Proximate analysis, developed at the Weende Experiment Station (Henneberg and Stohmann, 1860s), partitions a food into six fractions: moisture, crude protein, crude fat (ether extract), total ash, crude fibre, and nitrogen-free extract (NFE, the digestible carbohydrate estimated by difference). The word crude signals that each fraction is defined by the method, not by chemistry: crude protein is nitrogen times a factor, crude fat is whatever the solvent dissolves, crude fibre is whatever survives acid and alkali. These are the methods cited in the FSSAI Manual of Methods of Analysis of Foods and in AOAC Official Methods, and they underpin nutrition labelling under the Labelling and Display Regulations, 2020, as well as standards of identity (for example minimum protein in milk powder, maximum moisture in spices and cereals).

NFE (percent) = 100 − (moisture + crude protein + crude fat + total ash + crude fibre), so every error in the five measured fractions accumulates in NFE.

5.1 Moisture

Moisture controls microbial stability (through water activity), texture, economic value (water sold at the price of food) and compliance with standards (for example not more than 14 percent in wheat flour, 12 percent in most spice powders, 5 percent in milk powder).

Method	Conditions	Principle	Suitable for	Limitations
Air oven (AOAC 925.10, 930.15; IS 1155)	100 to 105 °C to constant weight (usually 3 to 5 hours, or 130 °C for 1 hour for flour); sample 2 to 5 g in a tared aluminium dish, cooled in a desiccator before weighing	Loss on drying; moisture = (initial − final weight) divided by initial weight × 100	Cereals, flours, pulses, spices, low-fat foods with little volatile matter	Overestimates where volatile oils, alcohol or acetic acid are lost; underestimates where oxidation adds weight; sugars caramelise and decompose, and bound water is not fully released; crust formation traps water
Vacuum oven (AOAC 934.06)	60 to 70 °C at a pressure not above 100 mm Hg (about 13 kPa) for 6 hours or to constant weight	Lower boiling point of water at reduced pressure allows drying without thermal decomposition	High-sugar foods (honey, jams, dried fruit, confectionery, fruit juices), foods prone to Maillard browning	Slow; requires vacuum pump and air bleed through sulfuric acid
Distillation, Dean and Stark (AOAC 925.04)	Sample refluxed with toluene or xylene; water co-distils and separates in a graduated receiver	Azeotropic distillation; water volume read directly	Spices and foods rich in volatile oils (where oven drying would lose the oil as moisture), cheese, fatty foods	Reads only to about 0.1 mL; emulsions and dissolved water in solvent cause small errors

Method	Conditions	Principle	Suitable for	Limitations
Karl Fischer titration (AOAC 977.10)	Titration of the sample (dissolved or extracted in methanol) with Karl Fischer reagent to a bipotentiometric end point	Water-specific reaction of iodine with sulfur dioxide in the presence of a base and methanol: I2 + SO2 + 3Base + CH3OH + H2O → 3Base·HI + Base·HSO4CH3 (one mole of iodine per mole of water); the base is now imidazole (pyridine-free reagents)	Low-moisture foods (dried milk, oils and fats, chocolate, dried vegetables, sugar, coffee, flour), where loss-on-drying methods are imprecise; the only method that measures water specifically rather than total volatiles	Aldehydes and ketones react with methanol to form water (special reagents needed); strong oxidants and reductants interfere; sample must release water to the solvent
Coulometric Karl Fischer	Iodine generated electrolytically at the anode; 1 mg of water corresponds to 10.71 coulombs by Faraday's law	Trace water, 10 µg to 10 mg, in oils, solvents, packaging films	Not for high-moisture samples	
Rapid methods: infrared moisture balance, microwave drying, near-infrared reflectance, nuclear magnetic resonance, capacitance grain meters	Minutes	Indirect; must be calibrated against the official oven method	Process control, grain procurement	Not official for enforcement

Moisture (percent) = [(W1 − W2) ÷ (W1 − W0)] × 100, where W0 is the empty dish, W1 the dish with sample before drying, W2 after drying.

5.2 Crude protein: the Kjeldahl method

Introduced by Johan Kjeldahl at the Carlsberg Laboratory (1883) and still the AOAC reference method (AOAC 955.04, 991.20 for milk, 2001.11; IS 7219). It measures total nitrogen (organic nitrogen plus ammonium, but not nitrate, nitrite, azo or nitro nitrogen without modification) and converts it to protein by a factor.

Step 1: Digestion (oxidation of organic matter and conversion of nitrogen to ammonium sulfate)

About 1 g of sample is heated with 15 to 25 mL of concentrated sulfuric acid, 10 to 15 g of potassium sulfate and a catalyst (0.3 to 1 g of copper sulfate, or copper with titanium dioxide, or selenium; mercuric oxide was the original catalyst but is now avoided for toxicity and because it binds ammonia, needing thiosulfate release) in a long-necked Kjeldahl flask or block digester at 380 to 420 °C for 1 to 3 hours, until the black charred liquid becomes clear and pale blue-green.

Organic nitrogen (in protein, R-NH2) + H2SO4 (with catalyst, heat) → (NH4)2SO4 + CO2 + SO2 + H2O

Roles of the reagents: sulfuric acid oxidises carbon and hydrogen and converts amino nitrogen to ammonium; potassium sulfate raises the boiling point of the acid from about 338 °C to about 380 to 400 °C, speeding digestion (too much salt and the temperature exceeds 410 °C, at which ammonia is lost); the copper catalyst accelerates oxidation of refractory nitrogen compounds (heterocyclic nitrogen in tryptophan, histidine, nicotinic acid digests slowly and incompletely without it).

Step 2: Neutralisation and steam distillation (liberation of ammonia)

The cooled digest is diluted with water and made strongly alkaline with 40 to 50 percent sodium hydroxide (excess, since the acid must be fully neutralised and the ammonium converted to free ammonia), then steam distilled.

$(NH_4)_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2NH_3\,(gas) + 2H_2O$

The ammonia is absorbed in a receiving flask containing 4 percent boric acid (about 25 to 50 mL) with a mixed indicator (methyl red and bromocresol green, or methyl red and methylene blue). Ammonia forms the ammonium-borate ion pair:

$NH_3 + H_3BO_3 \rightarrow NH_4^+ + H_2BO_3^-$

The indicator turns from pinkish-red (acid) to green as ammonia is absorbed. Boric acid is a weak acid (pKa 9.2) that holds the ammonia but does not itself consume the titrant, which is the elegance of the method: no standardised receiving acid is needed and there is no back-titration. The older alternative collects ammonia in a measured excess of standard sulfuric or hydrochloric acid and back-titrates the unreacted acid with standard sodium hydroxide.

Step 3: Titration

The borate formed is titrated with standard 0.1 N hydrochloric acid (or sulfuric acid) until the indicator returns from green to the original pink end point:

$H_2BO_3^- + H^+\,(\text{from HCl}) \rightarrow H_3BO_3$

One mole of hydrochloric acid corresponds to one mole of ammonia and therefore to one mole of nitrogen (14.007 g).

Calculation

Nitrogen (percent) = [(V sample − V blank) × N × 1.4007] ÷ W

where V is the titre in mL, N the normality of the acid, 1.4007 the milliequivalent weight of nitrogen (14.007) multiplied by 100 and divided by 1000, and W the sample weight in grams.

Crude protein (percent) = Nitrogen (percent) × F

Worked example: 1.000 g of wheat flour, titre 10.50 mL of 0.1 N HCl, blank 0.20 mL. Nitrogen = (10.50 − 0.20) × 0.1 × 1.4007 ÷ 1.000 = 1.443 percent. Protein with the general factor = 1.443 × 6.25 = 9.02 percent; with the wheat factor = 1.443 × 5.70 = 8.22 percent. The difference of 0.8 percentage points is why FSSAI standards specify the factor.

Nitrogen-to-protein conversion factors

Food	Factor	Basis
General (meat, eggs, pulses, mixed foods)	6.25	Assumes proteins contain 16 percent nitrogen: 100 ÷ 16 = 6.25
Milk and milk products	6.38	Milk proteins (casein, whey) contain about 15.67 percent nitrogen: 100 ÷ 15.67 = 6.38
Wheat and wheat products (whole wheat, flour, bread)	5.70	Wheat proteins (gliadin, glutenin) contain about 17.5 percent nitrogen because of high glutamine and proline content: 100 ÷ 17.54 = 5.70
Rice	5.95	Rice proteins contain about 16.8 percent nitrogen

Food	Factor	Basis
Soybean	5.71	High glutamine content
Gelatin	5.55	Glycine-rich
Almonds	5.18; other nuts and seeds 5.30 to 5.46	Arginine-rich storage proteins
Maize, barley, oats	6.25 (maize), 5.83 (barley, oats, rye)	Prolamin content

HPSC EXAM TRAP: Kjeldahl measures nitrogen, not protein
Any nitrogen source raises the apparent protein: urea (46.6 percent N), ammonium salts, and melamine (66.6 percent N), the adulterant of the 2008 Chinese infant formula and pet food scandals. FSSAI therefore detects urea in milk by the dimethylaminobenzaldehyde (DMAB) or urease test and melamine by LC-MS/MS; the Dumas combustion method (AOAC 992.15; sample burnt at 900 to 1000 °C in oxygen, nitrogen oxides reduced to N2 over copper and measured by a thermal conductivity detector) is a faster, reagent-free alternative that is equally fooled by non-protein nitrogen. True protein requires precipitation (trichloroacetic acid or copper hydroxide) before nitrogen determination, or amino acid analysis. Note also that the factor for milk (6.38) is higher than 6.25 and the factor for wheat (5.70) is lower; reversing them is a common MCQ trap.

5.3 Crude fat: Soxhlet solvent extraction

Devised by Franz von Soxhlet (1879) for milk fat and adopted as AOAC 920.39 (cereals), 960.39 (meat), 948.22 (nuts); IS 4684.

Apparatus and operation

The apparatus has three parts: a boiling flask containing the solvent, a central extraction chamber holding a cellulose thimble with 2 to 5 g of dried, finely ground sample (plugged with fat-free cotton), and a reflux condenser above. Solvent vapour rises through the side tube, condenses and drips onto the sample; the chamber fills with solvent, dissolving lipid; when the liquid reaches the top of the siphon arm, the entire contents siphon back into the flask. Each cycle (5 to 6 per hour) bathes the sample in fresh, distilled solvent, while the dissolved fat accumulates in the flask. This semi-continuous extraction runs for 6 to 16 hours (4 hours at 5 to 6 drops per second for most foods). The solvent is then recovered by distillation, the flask dried at 100 to 105 °C for 1 hour, cooled in a desiccator and weighed.

Crude fat (percent) = [(W flask with fat − W empty flask) ÷ W sample] × 100

Solvent choice

Solvent	Boiling range	Advantages	Disadvantages
Petroleum ether (light petroleum, hexane fraction)	40 to 60 °C	Cheaper, non-hygroscopic, does not dissolve sugars or water-soluble material, lower peroxide risk, more selective for true lipids (triglycerides, free fatty acids, sterols, carotenoids, waxes); the solvent prescribed in most FSSAI and AOAC food methods	Does not extract polar lipids (phospholipids, glycolipids)

Solvent	Boiling range	Advantages	Disadvantages
Diethyl ether	34.6 °C	Extracts a wider range including some phospholipids; historical Weende solvent (hence ether extract)	Absorbs up to 2 percent water, which then co-extracts sugars and urea, inflating results; forms explosive peroxides on storage; very flammable; hygroscopic
Hexane, chloroform-methanol (Folch and Bligh-Dyer)	69 °C; mixed	Chloroform-methanol gives total lipid including bound and polar lipids for research and labelling	Toxic solvents, not routine

Conditions and limitations

- The sample must be dried first (moisture blocks penetration of the non-polar solvent and water would be co-extracted) and finely ground to expose lipid; cereals and oilseeds are usually ground to pass a 1 mm sieve.
- Soxhlet measures free lipid only. Bound lipids in bakery products, dairy products and processed foods (lipoproteins, starch-lipid complexes, fat enclosed in intact cells) are released by acid hydrolysis before extraction (Weibull-Stoldt, AOAC 922.06: boiling with 3 N hydrochloric acid, filtering, drying, then Soxhlet), or by the Rose-Gottlieb alkaline (ammonia) ethanol-ether method for milk and infant formula (AOAC 905.02, Mojonnier flask), or the Gerber butyrometer method for milk (sulfuric acid digests protein, amyl alcohol aids separation, centrifugation reads fat directly; the field method of the Food Safety Officer).
- Automated Soxtec (submersion then rinsing) and accelerated solvent extraction reduce time to under one hour.
- Thermally labile lipids can oxidise during long extraction; the result is reported as crude fat or ether extract, never as triglyceride content.

5.4 Total ash and acid-insoluble ash

Total ash (dry ashing, AOAC 923.03, 942.05; IS 1155)

Ash is the inorganic residue after complete combustion of organic matter and is a measure of total mineral content (sodium, potassium, calcium, magnesium, phosphorus, iron, trace metals present as oxides, carbonates, phosphates, chlorides, sulfates). Procedure: 2 to 5 g of sample in a tared silica, porcelain or platinum crucible is first charred gently over a flame or hot plate until smoking ceases (to prevent loss by spattering and foaming), then ignited in a muffle furnace at 550 °C (a range of 525 to 600 °C is specified by different methods) for 4 to 6 hours or overnight, until the ash is white or light grey and free of carbon. The crucible is cooled in a desiccator and weighed to constant weight.

Total ash (percent) = [(W crucible with ash − W empty crucible) ÷ W sample] × 100

Why 550 °C: high enough to burn carbon completely within hours, low enough to limit volatilisation of chlorides, sodium, potassium, phosphorus, and of mercury, arsenic, selenium, cadmium and lead, and to limit fusion of the ash with the crucible. Black specks of unburnt carbon are treated by wetting the ash with water or dilute nitric acid, drying and re-ashing. Where sodium chloride is high (pickles, butter), 500 °C is used. Wet ashing (nitric, sulfuric and perchloric acids) and microwave digestion are preferred when the ash is to be analysed for individual metals (Part 4B). Sulfated ash (treatment with sulfuric acid before ignition, converting all cations to stable sulfates) is used for food additives and colours.

Acid-insoluble ash (AIA)

The total ash is boiled with 25 mL of dilute hydrochloric acid (about 10 percent, 1 part concentrated acid to 2.5 parts water) for 5 minutes; the soluble mineral salts dissolve, the solution is filtered through ashless filter paper, the residue is washed free of chloride with hot water, and the paper with residue is ignited again at 550 °C and weighed. The acid-insoluble ash is almost entirely silica: sand, soil, dust and grit. It is the statutory index of extraneous mineral adulteration in spices, flours and tea, and of poor washing of grains.

Commodity (FSS Food Products Standards and Food Additives Regulations, 2011)	Total ash maximum, percent	Acid-insoluble ash maximum, percent
Atta (whole wheat flour)	2.0 (on dry basis)	0.1
Maida (refined wheat flour)	1.0	0.1
Besan (Bengal gram flour)	5.0	0.5
Turmeric powder	9.0	1.5
Chilli powder	8.0	1.25
Coriander powder	9.0	1.5
Black pepper (whole)	7.0	1.5
Tea	4.0 to 8.0 (with water-soluble ash not less than 45 percent of total ash)	1.0
Milk powder	not more than 9.3 (whole), 11 (skimmed)	not specified; ash is a purity index for dairy

Related ash indices: water-soluble ash and alkalinity of water-soluble ash (tea, where low values indicate exhausted tea leaves), ash insoluble in water, and the ash content of sugar and honey as purity measures.

5.5 Crude fibre: the Weende method

Crude fibre is the organic residue that resists sequential boiling in dilute acid (simulating gastric digestion) and dilute alkali (simulating intestinal digestion) and is then lost on ignition (AOAC 962.09, 978.10; IS 10226).

Procedure

1. Weigh about 2 g of dried, fat-free sample (the Soxhlet residue is ideal; fat above 1 percent must be removed or it coats particles and interferes).
2. Acid digestion: boil gently with 200 mL of 1.25 percent (0.255 N) sulfuric acid under reflux for exactly 30 minutes, keeping the volume constant with hot water; filter through a linen cloth, Buchner funnel or fritted crucible, and wash with boiling water until the washings are no longer acid.
3. Alkali digestion: transfer the residue to 200 mL of 1.25 percent (0.313 N) sodium hydroxide and boil for exactly 30 minutes; filter and wash with boiling water until free of alkali, then with 25 mL of 1.25 percent sulfuric acid, three times with water, and finally with 25 mL of ethanol (and petroleum ether if fat remains).
4. Dry the residue in a crucible at 130 °C for 2 hours (or 100 to 105 °C overnight), cool in a desiccator and weigh (W1).
5. Ignite in a muffle furnace at 550 °C for 30 minutes (or until carbon free), cool and weigh (W2).

Crude fibre (percent) = [(W1 − W2) ÷ W sample] × 100

The loss on ignition is the crude fibre; the ash remaining (W2) is subtracted so that mineral matter trapped in the residue is not counted as fibre. Exact timing, acid and alkali concentrations and boiling intensity must be standardised because the result is purely operational.

What crude fibre is and is not

Crude fibre recovers 50 to 80 percent of the cellulose, only 10 to 50 percent of the lignin and under 20 percent of the hemicellulose, and none of the pectins, gums, beta-glucans or resistant starch; it therefore underestimates total dietary fibre by a factor of 2 to 5, and the ratio varies with the food. Crude fibre remains a legal standard of identity for spices, flours and animal feeds (for example FSSAI maximum crude fibre values in chilli powder and coriander powder, bran content in atta), but the nutrition label value for dietary fibre must come from the enzymatic-gravimetric method (AOAC 985.29 or 991.43, the Prosky method: digestion with heat-stable alpha-amylase, protease and amyloglucosidase, precipitation of soluble fibre with four volumes of ethanol, weighing of the residue corrected for its protein and ash) or AOAC 2009.01 and 2011.25 for total dietary fibre including resistant starch and low-molecular-weight oligosaccharides. For feeds, the Van Soest detergent system (neutral detergent fibre, acid detergent fibre, acid detergent lignin) is used.

5.6 Summary table of the proximate scheme

Fraction	Official principle	Key reagents and conditions	Reports	Classic error
Moisture	Loss on drying or Karl Fischer	100 to 105 °C air oven; 70 °C vacuum oven for sugary foods; iodine-sulfur dioxide-base-methanol reagent	Water and volatiles (oven) or water only (Karl Fischer)	Volatile oils counted as moisture; sugar decomposition
Crude protein	Kjeldahl: digestion, distillation, titration	Concentrated sulfuric acid, potassium sulfate, copper sulfate; 40 percent sodium hydroxide; 4 percent boric acid with mixed indicator; standard 0.1 N hydrochloric acid; factor 6.25 (6.38 milk, 5.70 wheat)	Total nitrogen × factor	Non-protein nitrogen (urea, melamine) counted as protein; wrong factor
Crude fat	Soxhlet continuous solvent extraction	Petroleum ether 40 to 60 °C (or diethyl ether), 4 to 16 hours, dried sample	Free lipid soluble in the solvent	Bound lipid missed; moisture in sample co-extracts sugars
Total ash	Dry ashing	Muffle furnace 550 °C to white or grey ash	Total minerals as oxides and salts	Loss of volatile elements and chlorides at higher temperatures; unburnt carbon
Acid-insoluble ash	Ash boiled in dilute hydrochloric acid, residue ignited	10 percent HCl, ashless filter paper, 550 °C	Silica, sand, dirt (adulteration index)	Incomplete washing of chloride

Fraction	Official principle	Key reagents and conditions	Reports	Classic error
Crude fibre	Weende sequential digestion	1.25 percent H2SO4 then 1.25 percent NaOH, 30 minutes each, dry at 130 °C, loss on ignition at 550 °C	Part of cellulose and lignin	Mistaken for dietary fibre on labels
Nitrogen-free extract	By difference	None	Digestible carbohydrate (starch, sugars) plus all accumulated errors	Not a measured quantity

HPSC EXAM TRAP: the numbers in the proximate scheme
Ash is 550 °C, not 950 °C. Crude fibre uses 1.25 percent acid and 1.25 percent alkali for 30 minutes each, acid first. Boric acid in Kjeldahl is 4 percent and the titrant is standard acid (not alkali, which is used only in the back-titration variant). Petroleum ether boils at 40 to 60 °C; the vacuum oven for sugary foods runs at 70 °C, not 105 °C. Karl Fischer is specific for water; the oven measures total volatiles.

Part 6. Stage 1 Revision Bank (Count-Based Pairs)

PART 6: STAGE 1 REVISION BANK, COUNT-BASED PAIR QUESTIONS

Method for every question: assign c1, c2, c3 = 1 for a correct statement and 0 for an incorrect statement; compute N = c1 + c2 + c3; map N = 0 to None, N = 1 to Only one, N = 2 to Only two, N = 3 to All three. Never pick an option until N is written down. With 0.25 negative marking, a guessed count question has an expected value of zero; a verified one is a full mark.

Question 1: Packaging polymers and retort pouches

Consider the following statements:

1. High-density polyethylene has a lower density but a higher degree of crystallinity than low-density polyethylene, which is why it is a better moisture barrier.
2. In a four-ply retort pouch the biaxially oriented nylon (OPA) layer is positioned between the aluminium foil and the inner cast polypropylene sealant layer to provide puncture and flex-crack resistance.
3. Ethylene vinyl alcohol copolymer retains its exceptional oxygen barrier at all relative humidities and is therefore used as a monolayer retort film.

How many of the above statements are correct?
(a) Only one (b) Only two (c) All three (d) None

Step-by-step verification:

- Statement 1: HDPE has higher crystallinity (70 to 90 percent versus 40 to 50 percent) and therefore higher density (0.941 to 0.965 g/cm³ versus 0.910 to 0.925 g/cm³), not lower; the barrier conclusion is right but the density clause is wrong, so the statement as a whole is wrong. c1 = 0.
- Statement 2: The four-ply construction is PET (12 µm) / aluminium foil (9 to 12 µm) / OPA (15 µm) / CPP (70 to 100 µm); nylon sits between foil and sealant to protect the foil from cracking. c2 = 1.
- Statement 3: EVOH is hygroscopic; above about 75 percent RH its oxygen transmission rate rises ten to a hundred fold and it suffers retort shock; it is always buried as a core layer between polyolefin skins and never used as a monolayer. The absolute qualifier at all relative humidities is the trap. c3 = 0.

N = c1 + c2 + c3 = 0 + 1 + 0 = 1. Answer: (a) Only one.

Question 2: Active and intelligent packaging systems

Consider the following statements:

1. A methylene blue oxygen indicator inside a pack is pink in the absence of oxygen and turns blue when oxygen enters through a leaking seal.
2. A potassium permanganate ethylene absorber changes colour from purple to brown as permanganate is reduced to manganese dioxide, indicating exhaustion.
3. A polymerisation-based time-temperature indicator such as the Lifelines Fresh-Check label is classified as active packaging because it reacts chemically with temperature.

How many of the above statements are correct?
(a) Only one (b) Only two (c) All three (d) None

Step-by-step verification:

- Statement 1: In the reduced (leuco) state in an oxygen-free pack the tablet is pink; oxygen re-oxidises methylene blue to its blue form. c1 = 1.
- Statement 2: KMnO4 (purple) oxidises ethylene and is itself reduced to MnO2 (brown); the colour change signals that capacity is spent. c2 = 1.

- Statement 3: A TTI monitors and communicates the temperature history; it does not act on the food or the headspace, so it is intelligent packaging regardless of the chemistry inside the label. The reasoning (reacts chemically) is a distractor. c3 = 0.

N = 1 + 1 + 0 = 2. Answer: (b) Only two.

Question 3: FSSAI migration regulations

Consider the following statements with reference to the Food Safety and Standards (Packaging) Regulations, 2018:

1. The overall migration limit for plastic food contact materials is 60 mg/kg of food or simulant, equivalently 10 mg/dm² of contact surface.
2. Food simulant B, 3 percent (w/v) acetic acid in water, represents acidic aqueous foods with a pH below 4.5 such as fruit juices, pickles and tomato products.
3. Newspapers and similar printed materials are prohibited for storing, wrapping or packaging food.

How many of the above statements are correct?
(a) Only one (b) Only two (c) All three (d) None

Step-by-step verification:

- Statement 1: Schedule IV sets the overall migration limit at 60 mg/kg or 10 mg/dm², identical to EU Regulation 10/2011; the two figures are linked by the convention of 6 dm² of packaging per kilogram of food (10 × 6 = 60). c1 = 1.
- Statement 2: Simulant B is the only acidic simulant, 3 percent acetic acid, assigned to foods below pH 4.5. c2 = 1.
- Statement 3: Regulation 4 of the 2018 Regulations prohibits newspaper and similar materials for storing and wrapping food, and also requires that the printed surface of any packaging must not come into direct contact with food; FSSAI had issued the same advisory in 2016. c3 = 1.

N = 1 + 1 + 1 = 3. Answer: (c) All three.

Question 4: Instrumental analysis and proximate methods

Consider the following statements:

1. Flame atomic absorption spectrometry achieves lower (better) detection limits than graphite furnace atomic absorption spectrometry because the flame atomises the whole sample continuously.
2. The Kjeldahl nitrogen-to-protein conversion factor for milk and milk products is 5.70.
3. The electron capture detector gives a response proportional to the number of carbon atoms in every organic compound and is therefore the universal detector for gas chromatography.

How many of the above statements are correct?
(a) Only one (b) Only two (c) All three (d) None

Step-by-step verification:

- Statement 1: The reverse is true: the flame wastes 90 to 95 percent of the sample in the spray chamber and atoms reside only milliseconds in the beam, so FAAS works at mg/L; the graphite furnace atomises the whole 10 to 50 µL aliquot into a confined volume and reaches µg/L, 100 to 1000 times better. c1 = 0.
- Statement 2: The milk factor is 6.38 (15.67 percent nitrogen in milk proteins); 5.70 is the wheat factor. c2 = 0.
- Statement 3: Equal-carbon response and universality describe the flame ionisation detector; the electron capture detector with its 63Ni beta source responds selectively to electronegative, chiefly halogenated, compounds and barely responds to hydrocarbons. c3 = 0.

N = 0 + 0 + 0 = 0. Answer: (d) None.

Absolute-qualifier mini-bank (mark each true or false before reading the key)

1. Aluminium foil is always a 100 percent barrier. False: below 15 µm pinholes make it a barrier only within a laminate.
2. All recycled plastics are banned for food contact in India. False: certified food-grade recycled PET from approved recyclers is permitted after the amendment framework that followed the 2018 Regulations.
3. Passing the overall migration test proves the material is toxicologically safe. False: specific migration limits must also be met.
4. Mercury is determined by hydride generation AAS. False: cold vapour AAS at 253.7 nm.
5. Only Simulant E is a solid. True: Tenax, poly(2,6-diphenyl-p-phenylene oxide), for dry foods.
6. C-enamel is never used for acid fruit packs. True: zinc oxide would be extracted; R-enamel or plain heavy-tin cans are used.
7. The refractive index detector can be used with gradient elution. False.
8. Crude fibre never equals dietary fibre. True in practice: crude fibre underestimates dietary fibre by a factor of 2 to 5.

Part 7. Stage 2 Descriptive Model Answers

PART 7: STAGE 2 DESCRIPTIVE MODEL ANSWERS

10-Mark Model Answer

Question: Explain the architecture, material composition, and advantages of retort pouches over conventional tinplate cans in food preservation.

Suggested marking scheme: definition and context 1 mark; layer-by-layer architecture with gauges and functions 3 marks; heat-penetration mechanism 2 marks; comparative advantages 2 marks; limitations and process requirements 1 mark; regulatory and quality-control linkage 1 mark. Target length: 400 to 450 words, one labelled schematic, one table.

Model answer

1. Definition and context

A retort pouch is a flexible, heat-sealable, hermetically closed laminate container that withstands thermal sterilisation at 115 to 135 °C (typically 121 °C for 20 to 40 minutes, delivering the required F0 value) so that low-acid foods become commercially sterile and shelf stable at ambient temperature. It was developed by the United States Army Natick Laboratories with Reynolds Metals and Continental Flexible Packaging in the 1960s and 1970s for military rations and is now used for ready-to-eat curries, rice, tuna, soups and pet food. It is the flexible-packaging alternative to the three-piece tinplate can developed from Appert and Durand (1810).

1. Architecture and material composition

Schematic (outside to inside, food contact at the bottom):

Layer	Material and gauge	Function
1. Outer	Biaxially oriented polyethylene terephthalate (BOPET), 12 µm	Reverse-printable surface, tensile and abrasion strength, heat resistance against sealing jaws
2. Barrier	Soft-annealed aluminium foil, 9 to 12 µm	Essentially total barrier to oxygen, water vapour, light and aroma once laminated (foil below 15 µm has pinholes on its own, bridged by the adjacent polymer layers); provides 18 to 36 months shelf life
3. Toughness (four-ply only)	Biaxially oriented nylon 6 (OPA), 15 µm	Puncture, flex-crack and abuse resistance; protects the foil in bony or sharp products
4. Inner sealant	Cast polypropylene (CPP), often ethylene-modified random copolymer, 70 to 100 µm	Food-contact layer; heat seals at about 160 °C, survives 121 to 135 °C without delamination or seal loss, inert and low taint; thick enough to bridge seal-area contamination

The three-ply pouch is PET / Al / CPP; the four-ply pouch is PET / Al / OPA / CPP. Layers are bonded by two-component polyurethane adhesives that must be fully cured before use to prevent migration of primary aromatic amines. Transparent, microwaveable variants replace foil with PET-SiOx, PET-AlOx or EVOH barrier layers at some sacrifice of shelf life.

1. Heat-penetration mechanism

Sterilisation of conduction-heating foods is unsteady-state conduction, and the time for the cold spot to reach process temperature scales with the square of the conduction distance. A 307 × 409 can has a cold-spot path of about 36 mm, whereas a filled pouch is a slab 15 to 25 mm thick with a half-thickness of 8 to 12 mm. The pouch therefore reaches the same F0 in 30 to 50 percent less time at the same retort temperature, and because the entire slab heats almost uniformly the surface layers are not overcooked while the centre catches up.

1. Advantages over the tinplate can

Aspect	Retort pouch	Tinplate can
Process time and energy	30 to 50 percent shorter; less steam, water and cooling	Long; surface overcooking
Nutrient and sensory quality	Better retention of thiamine, vitamin C, colour, texture, flavour	Lower
Weight and storage	Light; flat when empty, saving about 85 percent storage space	Heavy and bulky
Convenience	Tear-notch opening, boil-in-bag and microwave reheating (foil-free types)	Opener required; not microwaveable
Metal pickup	None (polymer contact)	Tin and iron pickup in plain cans (FSSAI tin limit 250 mg/kg); lacquer migration (bisphenol A from epoxy coatings)
Disposal	Lower mass, but multilayer laminate is hard to recycle	Readily recycled magnetically

1. Limitations and process requirements

The pouch must be processed in an overpressure retort (steam-air or water immersion) because internal vapour pressure at 121 °C would otherwise burst seals; pouches are racked flat to control thickness; residual air must be kept below about 10 mL by vacuum or steam-flush sealing; seals must be clean and are tested for strength, burst pressure and leaks; filling is slower and unit cost higher than canning; and the pouch needs a carton for distribution.

1. Regulatory linkage

The laminate materials must conform to Schedule IV of the FSS (Packaging) Regulations, 2018 (overall migration not more than 60 mg/kg or 10 mg/dm^2 by IS 9845, and specific migration limits for metals and DEHP), and the product must meet FSSAI commercial sterility requirements, verified by incubation at 37 °C for 10 days (and 55 °C for thermophiles). The Food Safety Officer inspects seal integrity, residual air, retort records (F0, come-up time, overpressure) and incubation results.

Conclusion: the retort pouch delivers can-equivalent sterility with a thinner geometry, giving faster, gentler processing and higher product quality, at the cost of stricter seal and process control.

15-Mark Model Answer

Question: Examine the statutory framework of Food Packaging under FSSAI Regulations (2018). Detail the food simulants used for migration testing, the difference between overall and specific migration limits, and the statutory prohibitions on packaging materials.

Suggested marking scheme: legal basis and structure of the Regulations 3 marks; general requirements 2 marks; food simulants with compositions and food classes 3 marks; overall versus specific migration with

the statutory numbers 3 marks; prohibitions 2 marks; enforcement, testing and conclusion 2 marks. Target length: 650 to 750 words, two tables, one callout for the number link.

Model answer

1. Legal basis and structure

Section 92 of the Food Safety and Standards Act, 2006 empowers the Food Safety and Standards Authority of India to make regulations, and Section 16 charges it with specifying standards for packaging. The Food Safety and Standards (Packaging) Regulations, 2018 were notified on 24 December 2018 and came into force on 1 July 2019, separating packaging from the earlier combined Packaging and Labelling Regulations, 2011 (labelling now sits in the Labelling and Display Regulations, 2020). The Regulations contain general requirements, specific requirements, and schedules: Schedule I (paper and board), Schedule II (glass), Schedule III (metals and alloys: tinplate, tin-free steel, aluminium, stainless steel), Schedule IV (plastics, with overall and specific migration limits and the Indian Standards for each polymer, for example IS 10146 polyethylene, IS 10910 polypropylene, IS 12252 PET, IS 10151 PVC, IS 12247 nylon, IS 13601 polycarbonate), and an indicative list of suggested packaging materials for food categories (dairy, fats and oils, cereals, meat and fish, fruits and vegetables, beverages, water, spices). The Regulations are to be read with the Contaminants, Toxins and Residues Regulations (limits for metals and migrants in food) and with the Plastic Waste Management Rules, 2016.

1. General requirements

- Packaging materials must be of food grade and must not transfer their constituents to food in quantities that could endanger human health, change the composition of the food unacceptably, or deteriorate its organoleptic properties (the inertness principle shared with EU Regulation 1935/2004).
- Materials must conform to the Indian Standards listed in the Schedules; the food business operator must obtain a certificate of conformity from an NABL-accredited laboratory.
- Packaging must be suitable for the type of food and its storage and processing conditions (hot fill, retort, frozen, microwave).
- Pigments and colourants in plastics must comply with IS 9833; the printed surface must not contact food.
- Containers must be free of defects: no rust, improper seaming, dents or leakage in cans; glass free of flaws that could cause breakage; paper free of harmful dyes and loading materials.

1. Food simulants for migration testing

Foods are too variable to test directly, so standardised liquids and solids that mimic the extractive power of food classes are used (EU Regulation 10/2011 Annex III; the same logic underlies IS 9845 referenced by Schedule IV).

Simulant	Composition	Food class represented
A	10 percent ethanol (v/v) in water	Aqueous foods, pH above 4.5 (milk-free, non-acidic beverages, vegetables)
B	3 percent acetic acid (w/v) in water	Acidic aqueous foods, pH below 4.5 (fruit juices, pickles, tomato products, soft drinks)
C	20 percent ethanol (v/v)	Alcoholic foods up to 20 percent alcohol; foods with lipophilic ingredients
D1	50 percent ethanol (v/v)	Alcoholic foods above 20 percent; oil-in-water emulsions; milk and dairy products
D2	Vegetable oil (refined olive or sunflower, less than 1 percent free fatty acids; substitutes 95 percent ethanol or iso-octane)	Fatty foods with free surface fat (oils, ghee, butter, fried snacks)

Simulant	Composition	Food class represented
E	Poly(2,6-diphenyl-p-phenylene oxide), Tenax, 60 to 80 mesh	Dry foods (cereals, rice, biscuits, powders)

Test conditions mirror intended use: 10 days at 40 °C for ambient storage, 2 hours at 70 °C for hot fill, 30 minutes at 121 °C for retort, 2 hours at 175 °C for ovenable trays. The material is exposed on one side at a defined surface-to-volume ratio and the simulant is analysed.

1. Overall migration versus specific migration

Criterion	Overall migration limit (OML)	Specific migration limit (SML)
Measures	Total non-volatile matter transferred, regardless of identity	A named substance (monomer, additive, catalyst residue, metal)
Basis	Inertness, a quality criterion	Toxicology: derived from tolerable daily intake
Method	Gravimetric residue after evaporating the simulant (IS 9845)	HPLC, GC-MS, LC-MS/MS, AAS or ICP-MS
Statutory value under Schedule IV	60 mg/kg of food or simulant, or 10 mg/dm² of contact surface	Barium 1 mg/kg; cobalt 0.05; copper 5; iron 48; lithium 0.6; manganese 0.6; zinc 25; antimony 0.04; arsenic, cadmium, chromium, lead and mercury 0.01 each; bis(2-ethylhexyl) phthalate (DEHP) 1.5

The two overall figures express one ceiling: regulators assume a 60 kg adult eats 1 kg of food per day that contacted 6 dm² of packaging (a 1 litre cube has 6 dm² of surface), so 10 mg/dm² × 6 dm²/kg = 60 mg/kg. The surface-based figure is applied to sheet materials and small containers; the mass-based figure to larger containers. Passing the overall test does not prove safety: a material can be under 60 mg/kg overall and still exceed the SML of a potent migrant such as bisphenol A or a primary aromatic amine, so both tests are required. Where a syllabus source quotes lead at 1 mg/kg, write the Schedule IV value (0.01 mg/kg) and note that 1 mg/kg is the barium limit.

1. Statutory prohibitions

- Newspaper and similar printed materials shall not be used for storing, wrapping or packaging food; printing inks carry lead, benzophenone, mineral oil hydrocarbons and primary aromatic amines, and the paper carries microbial load.
- The printed surface of any packaging material shall not come into direct contact with food (reverse printing or an unprinted inner layer is required).
- Recycled plastics shall not be used for packaging, storing, carrying or dispensing food, except that food-grade recycled PET produced by recyclers registered and certified under the subsequent FSSAI and Central Pollution Control Board framework (consistent with the amended Plastic Waste Management Rules) may be used for direct food contact; recycled paper and board bearing print or of unknown origin remain unsuitable for direct contact.
- Plastics not listed with an Indian Standard in Schedule IV, and pigments or colourants outside IS 9833, are not permitted; non-food-grade, rusted, dented or previously used containers, and lead-soldered cans, are prohibited.
- Packaging must not be used for a food category for which it is unsuitable (for example non-retortable films for sterilised products, or unlacquered cans for sulfur-rich foods).

1. Enforcement and testing linkage

The Food Safety Officer may draw samples of packaging along with food under Section 38 of the Act; overall migration is tested gravimetrically by IS 9845, metals by AAS or ICP-MS (lead and cadmium by graphite furnace or ICP-MS at 0.01 mg/kg), DEHP by GC-MS, bisphenol A by LC-MS/MS, and PET bottles for

packaged drinking water must meet BIS requirements including acetaldehyde control. Non-conformity attracts penalties under Sections 50 to 66 for sub-standard, unsafe or misbranded food.

Conclusion: the 2018 Regulations convert the inertness principle into enforceable numbers (60 mg/kg or 10 mg/dm² overall; substance-specific SMLs), standardised simulants, Indian Standard conformity, and clear prohibitions (newspaper, printed contact surfaces, uncertified recycled plastics), while the certified food-grade recycled PET exception shows the framework evolving towards circular packaging without compromising migration safety.